Geometry-Aware Safe Hierarchical Multi-Agent Reinforcement Learning for Multi-Spacecraft Close-Proximity Inspection

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Abstract

Autonomous close-proximity inspection with multiple inspector spacecraft requires coordinated task allocation, accurate six-degree-of-freedom maneuvering, and explicit safety handling near an inspected station. This paper develops a geometry-aware safe hierarchical multi-agent reinforcement learning framework for this problem. The framework first assigns inspection targets through a graph-based high-level policy, then tracks the assigned viewpoints with a low-level six-degree-of-freedom rigid-body controller, and finally projects the control commands through collision-avoidance and pointing constraints using safety filters based on control barrier functions. The resulting architecture couples learned coordination with model-based safety projection while preserving a modular separation between task allocation, continuous control, and safety filtering. Simulation studies in a MuJoCo/JAX environment evaluate coverage efficiency, coordination, and safety through ablations of graph representation, safety-aware control, and execution filtering.

Keywords: Multi-agent reinforcement learning, Spacecraft inspection, Hierarchical reinforcement learning, Control barrier functions, Safety-critical control, Graph neural networks

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1. Introduction

Sustained operation of large space infrastructure requires inspection capabilities that can identify surface anomalies, assess external components, and revisit regions of interest without exposing a crewed platform to unnecessary operational risk. The Lunar Gateway exemplifies a modular inspected station whose distributed pressurized elements, truss structures, radiators, and solar arrays create a geometrically heterogeneous close-proximity environment [1]. Deploying several inspector spacecraft can shorten mission duration and improve coverage, but it couples target allocation, six-degree-of-freedom maneuvering, sensor pointing, and collision avoidance.

Multi-spacecraft inspection is naturally hierarchical because assignment and motion evolve on different time scales. At the mission level, open inspection targets must be distributed among the available inspectors without duplicate service or persistently inefficient routes. At the execution level, each inspector must translate toward an assigned viewpoint while simultaneously orienting its sensor toward the corresponding station surface. The resulting motions must remain safe with respect to the nonconvex station geometry and the other inspectors, including during transient reassignment. A monolithic policy must therefore resolve a mixed discrete–continuous problem with long-horizon credit assignment and pointwise safety requirements, which can impede both training and interpretation.

Cooperative inspection research has established information-driven guidance [2], hierarchical learned viewpoint planning [3–6], and scalable or asynchronous multi-agent execution with run-time assurance [7, 8]. Model-based studies provide explicit multi-spacecraft safety around inspection targets [9–11], whereas learning-based six-degree-of-freedom methods address inspection, rendezvous, cooperative pose control, and constrained attitude tracking [12–16]. Image-based autonomy and uncertainty-aware proximity control further strengthen spacecraft operations [17–19], but none integrates nonduplicate surface assignment, multi-agent rigid-body control, and collision filtering at every step.

Safety cannot be represented reliably by mission reward alone because rare collisions may be hidden by high average return. Control barrier functions (CBFs) and high-order control barrier functions (HOCBFs) convert state constraints into online control inequalities [20, 21], while sampled-data analysis motivates faster barrier updates and explicit inter-sample margins [22]. Spacecraft studies combine reinforcement learning with barrier or run-time-assurance mechanisms for inspection, debris avoidance, attitude reorientation, and uncertain proximity operations [12, 23–25]. General safe multi-agent reinforcement learning (MARL) methods learn graph barriers or incorporate barrier information into policy optimization [26, 27], and hierarchical MARL with CBFs (HMARL-CBF) combines temporally abstract skills with a safety filter [28]; none of these methods addresses explicit inspector–surface assignment followed by full-pose station inspection.

Hierarchical reinforcement learning separates slow task decisions from fast control, while message-passing networks provide permutation-consistent relational context [29]. Holder et al. [30] apply multi-agent learning to sequential satellite assignment. Multi-agent proximal policy optimization (MAPPO), an extension of proximal policy optimization (PPO), provides centralized training with decentralized execution (CTDE) for cooperative control [31], whereas multi-agent guided policy optimization (MAGPO) introduces a guiding policy to improve coordination [32]. The unresolved integration question is whether graph assignment, learned multi-inspector rigid-body execution on the special Euclidean group SE(3), analytic safety projection, and detailed contact validation can operate in one randomized close-proximity mission.

This paper develops a geometry-aware safe hierarchical MARL framework for multi-spacecraft close-proximity inspection. A high-level graph neural network (GNN) represents inspector–target relations and predicts context-dependent assignment utilities, after which a masked sequential decoder enforces unique assignments among open targets. A shared decentralized actor executes the assigned goals under centralized Safe-MAPPO training, using full-pose observations and emitting station-frame acceleration and body-frame torque commands. Analytic CBF and HOCBF quadratic programs (QPs) finally project translational and attitude commands with respect to station geometry, inter-inspector separation, and surface-pointing requirements. This ordering preserves the distinction between learned mission decisions, learned continuous control, and model-based safety execution.

The main contributions are summarized as follows.

- A unified hierarchical formulation is established for randomized multi-spacecraft inspection, combining graph-structured target allocation, goal-conditioned SE(3) control, and explicit close-proximity safety filtering in one closed loop.

- A GNN allocator learns inspector–target utilities directly from relational mission observations without heuristic assignment labels, while a masked sequential decoder guarantees that concurrently selected open targets are distinct.

- A Safe-MAPPO execution policy couples centralized reward and safety-cost critics with decentralized full-pose control, and CBF/HOCBF filters address bounding-body station clearance, inter-inspector separation, and sensor pointing at the assigned station surface.

- A MuJoCo/JAX protocol evaluates randomized missions using task, control, barrier, and contact metrics in controlled multi-seed studies.

The remainder of this paper formulates the inspection problem in Section 2, presents the hierarchical method in Section 3, reports simulation results and their implications in Section 4, and concludes in Section 5.

2. Problem Formulation

2.1. Multi-Spacecraft Inspection Task

A close-proximity mission is formulated in a station-centered frame $\mathcal{F}_{\mathrm{S}}$ whose origin and axes are fixed to the inspected station during the maneuver. Let $\mathcal{N} = \{1, \ldots, N\}$ denote the inspectors and let an active set $\mathcal{M}$ of M targets be sampled from the candidate catalog $\mathcal{M}_{\mathrm{cat}}$ at reset. Each target is the viewpoint–surface pair

$$y_j = \left(p_j^v, p_j^s\right), \tag{1}$$

where $p_j^v \in \mathbb{R}^3$ specifies the inspector location and $p_j^s \in \mathbb{R}^3$ specifies the station surface observed at that location.

The body-to-station rotation $R_i(t)$ belongs to the special orthogonal group SO(3). For inspector position $p_i(t)$, body-to-station rotation $R_i(t)$, and body-frame sensor axis e_z^b, the sensor and line-of-sight directions are

$$b_i(t) = R_i(t)e_z^b, \qquad d_{ij}(t) = \frac{p_j^s - p_i(t)}{\left\|p_j^s - p_i(t)\right\|}. \tag{2}$$

Target j is inspected when one inspector simultaneously satisfies

$$\left\|p_i(t) - p_j^v\right\| \leq r_{\mathrm{vis}}, \qquad b_i(t)^\top d_{ij}(t) \geq a_{\mathrm{vis}}, \tag{3}$$

where r_{vis} and a_{vis} are the visit radius and minimum alignment. For visited flag $z_j(t) \in \{0, 1\}$, mission coverage is

$$C(t) = \frac{1}{M} \sum_{j \in \mathcal{M}} z_j(t). \tag{4}$$

Success requires $C(t) = 1$ before the horizon. At option index k, the high-level action $\sigma_i^k \in \mathcal{M} \cup \{\varnothing\}$ assigns inspector i to an open target or to an idle pose-hold state, subject to

$$\sigma_i^k \neq \sigma_\ell^k, \qquad i \neq \ell, \quad \sigma_i^k, \sigma_\ell^k \in \{j \mid z_j = 0\}. \tag{5}$$

The low-level controller receives $y_{\sigma_i^k}$ until the next option boundary; a completed goal is retained as a pose-hold reference, and $\varnothing$ holds the current pose. The objective is to complete coverage efficiently while satisfying the pointwise collision and pointing constraints below.

2.2. Inspector Spacecraft Dynamics on SE(3)

Each inspector is a fully actuated free rigid body with pose $g_i = (R_i, p_i) \in \mathrm{SE}(3)$ and state $x_i = (p_i, R_i, v_i, \omega_i)$, following the intrinsic spacecraft model in [33]. The translational variables p_i, v_i are expressed in $\mathcal{F}_{\mathrm{S}}$, whereas ω_i and torque are expressed in the body frame. The policy and safety layer use the mixed-coordinate input

$$u_i(t) = \left(a_i^{\mathrm{S}}(t), \tau_i^b(t)\right), \qquad a_i^{\mathrm{S}}(t), \tau_i^b(t) \in \mathbb{R}^3, \tag{6}$$

where the equivalent body-frame force is $f_i^b = m_i R_i^\top a_i^{\mathrm{S}}$. For mass m_i and body inertia J_i, the dynamics are

$$\begin{aligned}
\dot{p}_i &= v_i, \\
\dot{R}_i &= R_i \omega_i^\wedge, \\
m_i \dot{v}_i &= R_i f_i^b = m_i a_i^{\mathrm{S}}, \\
J_i \dot{\omega}_i &= \tau_i^b - \omega_i^\wedge J_i \omega_i,
\end{aligned} \tag{7}$$

where $a^\wedge b = a \times b$. The policy, safety filter, and evaluation all use Eq. (7), so translation and attitude-dependent pointing remain coupled through the same full-pose state.

2.3. Coverage, Collision Avoidance, and Pointing Constraints

Coverage follows Eq. (3), while collision avoidance uses inspector centers and an inflated station keep-out model. Let $\mathcal{O}_{\mathrm{S}} = \{1, \ldots, L\}$ index the station keep-out primitives after inflation by the inspector collision radius. For each primitive $\ell \in \mathcal{O}_{\mathrm{S}}$, positive $d_\ell^{\mathrm{S}}(p)$ denotes exterior signed clearance. For a capsule with segment endpoints a_ℓ, b_ℓ and inflated radius r_ℓ,

$$\begin{aligned} d_\ell^{\mathrm{S}}(p) &= \|p - \Pi_\ell(p)\|_2 - r_\ell, \\ \Pi_\ell(p) &= a_\ell + \alpha_\ell(p)(b_\ell - a_\ell), \\ \alpha_\ell(p) &= \mathrm{clip}_{[0,1]}\left(\frac{(p - a_\ell)^\top (b_\ell - a_\ell)}{\|b_\ell - a_\ell\|_2^2}\right). \end{aligned} \tag{8}$$

For an oriented bounding box (OBB) with center o_ℓ, rotation $A_\ell \in \mathrm{SO}(3)$, and inflated half extents e_ℓ,

$$q_\ell(p) = \left|A_\ell^\top (p - o_\ell)\right| - e_\ell, \tag{9}$$

and its signed clearance is

$$d_\ell^{\mathrm{S}}(p) = \|\max\left(q_\ell(p), 0\right)\|_2 + \min\left(\max_m q_{\ell,m}(p), 0\right), \tag{10}$$

where elementwise maximum is used in the first term, and $d_\ell^{\mathrm{S}}(p; m)$ denotes the same primitive enlarged by an additional margin $m \geq 0$. JAX differentiates the active closest-feature branch; feature transitions use the selected automatic-differentiation subgradient and are validated through solver diagnostics and MuJoCo contacts. The inspector-station safety constraints are

$$h_{i\ell}^{\mathrm{S}}(p_i) = d_\ell^{\mathrm{S}}(p_i) \geq 0, \qquad i \in \mathcal{N},\ \ell \in \mathcal{O}_{\mathrm{S}}. \tag{11}$$

This inflated-geometry representation allows the point p_i to conservatively protect the finite-size inspector body.

Inspector-inspector collision avoidance is imposed through pairwise separation constraints. For a minimum separation distance d_{safe}, the pairwise barriers are

$$h_{i\ell}^{\mathrm{I}}(p_i, p_\ell) = \|p_i - p_\ell\|^2 - d_{\mathrm{safe}}^2 \geq 0, \qquad i, \ell \in \mathcal{N},\ i < \ell. \tag{12}$$

Equations (11) and (12) define the translational safety set for all inspectors.

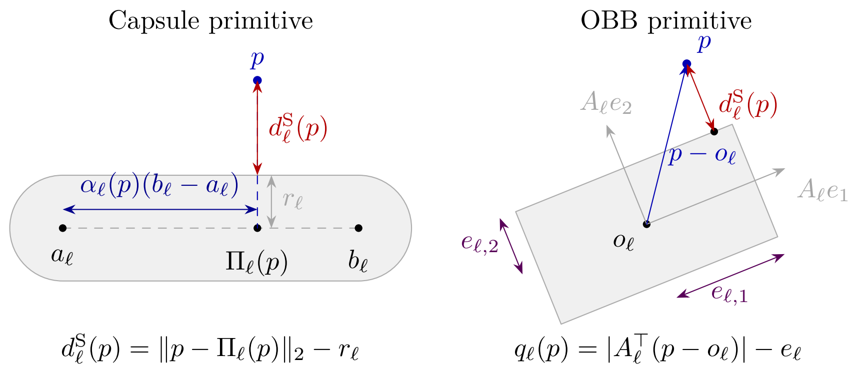

Figure 1: Signed-distance computation for the inflated station keep-out primitives. The capsule distance uses the closest point $\Pi_\ell(p)$ on the center segment, while the OBB distance first maps the query point into the box frame and then measures the excess vector $q_\ell(p)$ relative to the half extents e_ℓ.

The pointing constraint is defined with respect to the surface point associated with the assigned inspection target. If inspector i is assigned to target j, the instantaneous pointing margin is

$$h_{ij}^{\mathrm{P}}(p_i, R_i) = b_i^\top d_{ij} - a_{\mathrm{vis}}. \tag{13}$$

The surface-pointing requirement is active for inspection completion and according to the configured pointing-filter activation rule. Thus, an inspector must not only reach the viewpoint p_j^v, but also orient its sensor axis toward the corresponding station surface point p_j^s.

The feasible closed-loop behavior is therefore characterized by simultaneous satisfaction of the station keep-out constraints, pairwise separation constraints, and active surface-pointing constraints, while maximizing the coverage metric in Eq. (4).

3. Geometry-Aware Safe Hierarchical MARL Framework

3.1. Framework Overview

The proposed framework realizes the inspection problem as a hierarchical closed-loop policy with an explicit safety-critical execution layer. At each option update k, a graph-based allocator observes the current inspector states, visited target flags, and target geometry, and produces a nonconflicting assignment $\sigma^k = (\sigma_1^k, \ldots, \sigma_N^k)$ satisfying Eq. (5). Between option updates, each inspector follows a goal-conditioned low-level policy that maps its SE(3) state and assigned target geometry to a normalized six-dimensional action $\mathbf{a}_i(t) \in [-1,1]^6$. The action is mapped to a nominal station-frame acceleration and body-frame torque and then passed through a CBF/HOCBF control filter, which returns the executed mixed-coordinate input $u_i(t) = (a^{\mathrm{S}}_{i,\mathrm{safe}}, \tau^{b}_{i,\mathrm{safe}})$ used by the dynamics in Eq. (7) through the force mapping defined above. The resulting policy composition is

$$\sigma^k = \pi^{\mathrm{H}}_{\theta}(G^k), \quad \mathbf{a}_i(t) = \pi^{\mathrm{L}}_{\phi}\Big(o_i(t, \sigma^k_i)\Big), \quad u_i(t) = \mathcal{S}_{\mathrm{CBF}}\Big(\mathbf{a}_i(t), x(t), \sigma^k_i\Big), \quad (14)$$

where θ and ϕ denote the trainable parameters of the high-level assignment policy $\pi^{\mathrm{H}}_{\theta}$ and the low-level Safe-MAPPO policy π^{L}_{ϕ}, respectively, and $\mathcal{S}_{\mathrm{CBF}}$ denotes the safety filter induced by the station keep-out, pairwise separation, and surface-pointing constraints. The graph G^k encodes inspectors and inspection targets as nodes together with geometry-dependent edges and completion flags, while the open-target mask restricts the feasible assignments, and $o_i(t, \sigma^k_i)$ denotes the local goal-conditioned observation of inspector i. This architecture separates combinatorial coordination from continuous control while preserving a single SE(3) closed-loop dynamics model. The high-level module learns assignment utilities that affect service order and route efficiency, the masked decoder excludes concurrent duplicate assignments, the low-level module learns coordinated six-degree-of-freedom maneuvering, and the final filter applies pointwise constraints before the MuJoCo/JAX dynamics step. The framework is therefore safety aware at both training and execution: safety costs shape the learned policy, whereas CBF/HOCBF constraints act on the executed control input.

3.2. Graph-Based High-Level Task Allocation

The high-level module treats task allocation as a bipartite graph scoring problem between inspectors and the candidate target catalog, with an open-target mask enforcing assignment feasibility. Let $\mathcal{M}^k_{\mathrm{open}} = \{j \in \mathcal{M} \mid z_j(t_k) = 0\}$ denote the active unvisited targets at option update k. For every catalog entry, define the unavailable indicator $\bar{z}^k_j = \mathbb{I}\{j \notin \mathcal{M} \;\vee\; z_j(t_k) = 1\}$ and the closed-catalog fraction $\bar{C}^k = M^{-1}_{\mathrm{cat}} \sum_{j \in \mathcal{M}_{\mathrm{cat}}} \bar{z}^k_j$. For collision-feasible low-level guidance, let $\bar{p}^v_j$ denote the target viewpoint after a local signed-distance projection that places the inspector center outside the sampled-data station control envelope; the original p^v_j remains the point used by the visit condition in Eq. (3). For each inspector–target pair $(i,j) \in \mathcal{N} \times \mathcal{M}_{\mathrm{cat}}$, define $\rho^{g,k}_{ij} = \bar{p}^v_j - p_i(t_k)$, $\rho^{v,k}_{ij} = p^v_j - p_i(t_k)$, and the body-frame surface direction

$d_{ij}^{b,k} = R_i(t_k)^\top d_{ij}(t_k)$. The normalized sampled path clearance and current-assignment indicator are

$$
\begin{aligned}
c_{ij}^k &= \mathrm{clip}_{[-1,1]}\left(\frac{1}{c_\mathrm{S}} \min_{\lambda\in\Lambda} \min_{\ell\in\mathcal{O}_\mathrm{S}} d_\ell^\mathrm{S}\Big(p_i(t_k) + \lambda\rho_{ij}^{g,k}; m_\mathrm{s}^\mathrm{S}\Big)\right), \\
\iota_{ij}^k &= \mathbb{I}\Big\{\sigma_i^{k-1} = j\Big\},
\end{aligned}
\tag{15}
$$

where $c_\mathrm{S} > 0$ is a clearance normalization constant and $\Lambda = \{0, 1/8, \ldots, 1\}$ contains nine uniformly spaced samples along the direct guidance segment. The quantity m_s^S denotes the sampled-data station inflation introduced formally with the safety-update interval in Section 3.4. The 21-dimensional edge feature is

$$
\begin{aligned}
\xi_{ij}^k = \Bigg[&\rho_{ij}^{g,k}, \rho_{ij}^{v,k}, \left\|\rho_{ij}^{g,k}\right\|_2, \left\|\rho_{ij}^{v,k}\right\|_2, \bar{z}_j^k, \bar{C}^k, \\
&\frac{v_i(t_k)}{v_{\max}}, \frac{\omega_i(t_k)}{\omega_{\max}}, d_{ij}^{b,k}, c_{ij}^k, \iota_{ij}^k\Bigg] \in \mathbb{R}^{21},
\end{aligned}
\tag{16}
$$

where $v_{\max}$ and $\omega_{\max}$ are the translational- and angular-speed limits. The relative vectors retain directional geometry, their norms provide explicit radial cues, the velocity and angular-velocity terms expose the current rigid-body motion, and $d_{ij}^{b,k}$ encodes the attitude-dependent pointing geometry. The sampled clearance summarizes whether a direct option approaches the station envelope, while ι_{ij}^k identifies the currently retained edge and the minimum hold age is enforced separately by the fixed-edge assignment mask. The unavailable flag and closed-catalog fraction provide target-level and mission-progress context within the fixed-size dense graph, while the open-target mask excludes completed or inactive targets from the assignment policy.

The graph neural allocator combines an independently scored local edge with masked inspector and target context extracted from the dense bipartite edge tensor $\{\xi_{ij}^k\}_{i,j}$. The shared edge encoder and local branch are

$$
\begin{aligned}
e_{ij}^k &= \tanh\Big(W_\mathrm{e}\xi_{ij}^k + b_\mathrm{e}\Big), \\
\ell_{ij}^k &= \tanh\Big(W_\mathrm{l}e_{ij}^k + b_\mathrm{l}\Big).
\end{aligned}
\tag{17}
$$

Let $m_j^k = 1 - \bar{z}_j^k$ denote the open-target mask and $M_\mathrm{open}^k = \max(\sum_j m_j^k, 1)$ denote the protected open-target count. The inspector and target context vectors are

$$
\mu_i^k = \frac{1}{M_\mathrm{open}^k} \sum_{j\in\mathcal{M}_\mathrm{cat}} m_j^k e_{ij}^k, \qquad \nu_j^k = \frac{1}{N}\sum_{i\in\mathcal{N}} e_{ij}^k. \tag{18}
$$

The masked mean μ_i^k summarizes the open alternatives available to inspector i, whereas ν_j^k summarizes competition among inspectors for target j. The graph context and fused representation are

$$\begin{aligned} \mathbf{g}_{ij}^k &= \tanh\Big(W_{\mathrm{c}}[e_{ij}^k, \mu_i^k, \nu_j^k] + b_{\mathrm{c}}\Big), \\ f_{ij}^k &= \tanh\Big(W_{\mathrm{f}}[\ell_{ij}^k, \mathbf{g}_{ij}^k] + b_{\mathrm{f}}\Big), \\ \hat{s}_{ij}^k &= w_{\mathrm{loc}}^\top \ell_{ij}^k + b_{\mathrm{loc}} + w_{\mathrm{G}}^\top f_{ij}^k + b_{\mathrm{G}}. \end{aligned} \tag{19}$$

The graph-residual head $(w_{\mathrm{G}}, b_{\mathrm{G}})$ is initialized at zero, so optimization begins from a stable local edge scorer and learns team-context corrections only when they improve the closed-loop return. This residual construction prevents an untrained graph branch from immediately overwhelming useful pairwise geometry while still allowing every assignment utility to depend on competing inspectors and alternative open targets.

No additive distance score or heuristic assignment label is used, so the graph network must infer useful assignment preferences from the state and geometry features in Eq. (16). A deterministic diagnostic assignment can be obtained by a masked greedy decoder over the learned score matrix $\widehat{S}^k = [\hat{s}_{ij}^k]$. Starting from empty assigned inspector and target sets $\mathcal{I}_0 = \emptyset$ and $\mathcal{J}_0 = \emptyset$, the decoder repeats

$$(i_r, j_r) = \arg \max_{i \in \mathcal{N} \setminus \mathcal{I}_{r-1},\ j \in \mathcal{M}_{\mathrm{open}}^k \setminus \mathcal{J}_{r-1}} \hat{s}_{ij}^k, \tag{20}$$

and sets $\sigma_{i_r}^k = j_r$ until every inspector is assigned or no open target remains. The masks enforce the non-duplication constraint in Eq. (5), while the learned score determines which feasible assignment is preferred. The proposed high-level policy uses the same masks to sample unique inspector–target edges without replacement during training and execution. Equation (20) provides a deterministic decoding baseline. For a dense bipartite graph, the shared edge encoder and contextual aggregation require $\mathcal{O}(NM_{\mathrm{cat}})$ computation and storage per option, while the sequential masked decoder requires at most $\mathcal{O}(N^2 M_{\mathrm{cat}})$ score inspections. Figure 2 summarizes this graph-based allocation pipeline.

3.3. Safe-MAPPO for Low-Level SE(3) *Control*

Given the assigned target, the low-level policy outputs a normalized continuous action for each inspector. Compared with MAPPO [31], Safe-MAPPO augments the centralized reward critic with a safety-cost critic

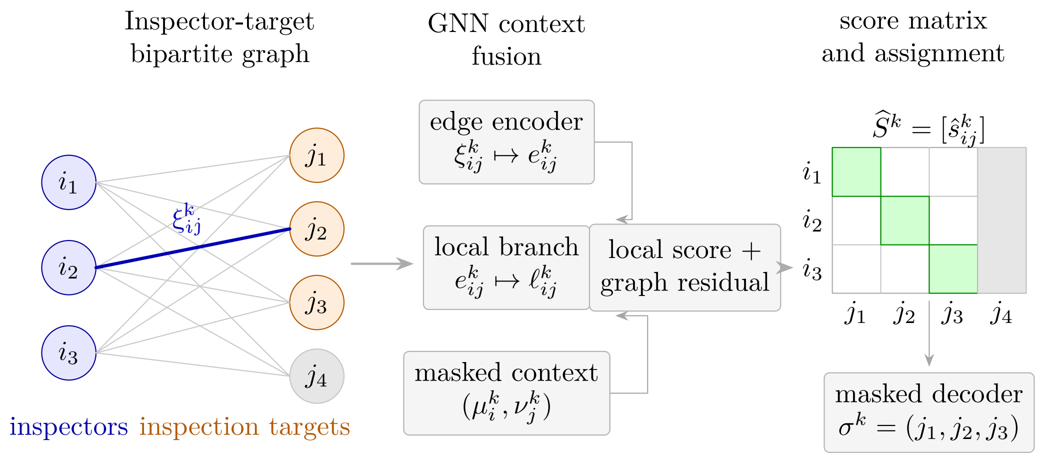

Figure 2: Graph-based high-level task allocation. A dense inspector–target bipartite graph is built from rigid-body state, task geometry, path clearance, and completion features; the GNN adds a masked team-context residual to each local edge utility, and a masked assignment operator selects nonduplicated open targets.

that estimates future collision and shield-intervention costs. Let n index the low-level control instants within high-level option k, let t_n denote the corresponding physical time, and let $j = \sigma_i^k$ denote the target assigned to inspector i, with sampled variables such as $v_{i,n}$ understood as $v_i(t_n)$. The goal-conditioned actor observation is

$$\begin{aligned} o_{i,n} = \Big[&p_i, v_i, \rho_{ij}^g, R_i^\top \rho_{ij}^g, \|\rho_{ij}^g\|_2, R_i^\top \rho_{ij}^s, \chi_{ij}, \\ &q_i, \omega_i, \Delta p_{i\kappa_i}, \Delta v_{i\kappa_i}, d_{i\kappa_i}, C, \tilde{c}_i^{\mathrm{S}}, d_{\mathrm{min}}^{\mathrm{I}}\Big] \in \mathbb{R}^{34}, \end{aligned} \tag{21}$$

where all quantities are evaluated at control instant n, $\rho_{ij}^g = \bar{p}_j^v - p_i$, $\rho_{ij}^s = p_j^s - p_i$, and $\chi_{ij} = b_i^\top d_{ij}$ is the surface-pointing alignment. The index κ_i identifies the nearest other inspector, with $\Delta p_{i\kappa_i} = p_i - p_{\kappa_i}$, $\Delta v_{i\kappa_i} = v_i - v_{\kappa_i}$, and $d_{i\kappa_i} = \|\Delta p_{i\kappa_i}\|_2$. The station-clearance observation is $\tilde{c}_i^{\mathrm{S}} = \min_{\ell \in \mathcal{O}_{\mathrm{S}}} d_\ell^{\mathrm{S}}(p_i; m_{\mathrm{s}}^{\mathrm{S}})$, the nominal bounding-body clearance is $c_i^{\mathrm{S}} = \min_{\ell \in \mathcal{O}_{\mathrm{S}}} d_\ell^{\mathrm{S}}(p_i; 0)$, and the team separation scalar is $d_{\mathrm{min}}^{\mathrm{I}} = \min_{i<\ell} \|p_i - p_\ell\|_2$. The sign-canonical unit quaternion q_i is used only as a compact numerical representation of R_i in the neural-network input, while the pose, dynamics, and control constraints remain defined on SE(3).

The homogeneous inspectors share a decentralized actor consisting of two fully connected hidden layers with hyperbolic-tangent activations. The actor defines a squashed Gaussian policy

$$
\begin{aligned}
\widetilde{\mathbf{a}}_{i,n} &\sim \mathcal{N}\Big(\mu_\phi(o_{i,n}), \mathrm{diag}\Big(e^{2s_\phi}\Big)\Big), \\
\mathbf{a}_{i,n} &= \tanh(\widetilde{\mathbf{a}}_{i,n}) = \begin{bmatrix} \mathbf{a}^{\mathrm{tr}}_{i,n} \\ \mathbf{a}^{\mathrm{rot}}_{i,n} \end{bmatrix},
\end{aligned} \tag{22}
$$

where $s_\phi \in \mathbb{R}^6$ is a learned log-standard-deviation vector shared across inspectors, and $\mathbf{a}^{\mathrm{tr}}_{i,n}, \mathbf{a}^{\mathrm{rot}}_{i,n} \in \mathbb{R}^3$ denote the normalized translational and rotational action components. The normalized action is converted into the nominal commands

$$
\bar{a}^{\mathrm{S}}_{i,n} = a_{\max}\mathbf{a}^{\mathrm{tr}}_{i,n}, \qquad \bar{\tau}^{b}_{i,n} = \tau_{\max}\mathbf{a}^{\mathrm{rot}}_{i,n}, \tag{23}
$$

where $\bar{a}^{\mathrm{S}}_{i,n}$ is the nominal translational acceleration in $\mathcal{F}_{\mathrm{S}}$ and $\bar{\tau}^{b}_{i,n}$ is the nominal body-frame torque. The safety layer in Section 3.4 filters both commands and maps the safe station-frame acceleration to the body-frame force $f^b_i = m_i R_i^\top a^{\mathrm{S}}_{i,\mathrm{safe}}$ required by Eq. (7).

Training follows the CTDE paradigm. The centralized observation is the concatenation

$$
o^{\mathrm{G}}_n = \mathrm{col}(o_{1,n}, \ldots, o_{N,n}) \in \mathbb{R}^{34N}. \tag{24}
$$

A shared critic encoder processes o^{G}_n and produces an agent-indexed reward-value head $V^r_{\vartheta,i}(o^{\mathrm{G}}_n)$ and safety-cost head $V^c_{\vartheta,i}(o^{\mathrm{G}}_n)$ for every inspector. The reward head estimates the discounted return of the inspection objective, whereas the cost head estimates the discounted accumulation of unsafe proximity and translational filter intervention. Only the shared actor $\pi^{\mathrm{L}}_\phi(\mathbf{a}_{i,n} \mid o_{i,n})$ is required during decentralized execution. Figure 3 summarizes the Safe-MAPPO training and execution pathways.

The task reward combines viewpoint acquisition, cooperative coverage, surface pointing, control efficiency, and safety-aware shaping. Define the assigned guidance-point distance and the open-target coverage potential as

$$
d^g_{i,n} = \|p_i(t_n) - \bar{p}^v_j\|_2, \qquad \Phi_n = \frac{1}{M}\sum_{j\in\mathcal{M}} (1 - z_j(t_n)) \min_{i\in\mathcal{N}} \|p_i(t_n) - \bar{p}^v_j\|_2. \tag{25}
$$

The progress terms are $\Delta d^g_{i,n} = d^g_{i,n} - d^g_{i,n+1}$ and $\Delta\Phi_n = \Phi_n - \Phi_{n+1}$, while the original viewpoint p^v_j remains the position used by the inspection event in Eq. (3). For any event $\mathcal{E}$, $\mathbb{I}\{\mathcal{E}\} = 1$ if $\mathcal{E}$ is true and 0 otherwise. The number of newly inspected targets is $N^{\mathrm{new}}_{n+1} = \sum_{j\in\mathcal{M}} (z_j(t_{n+1}) - z_j(t_n))$. The

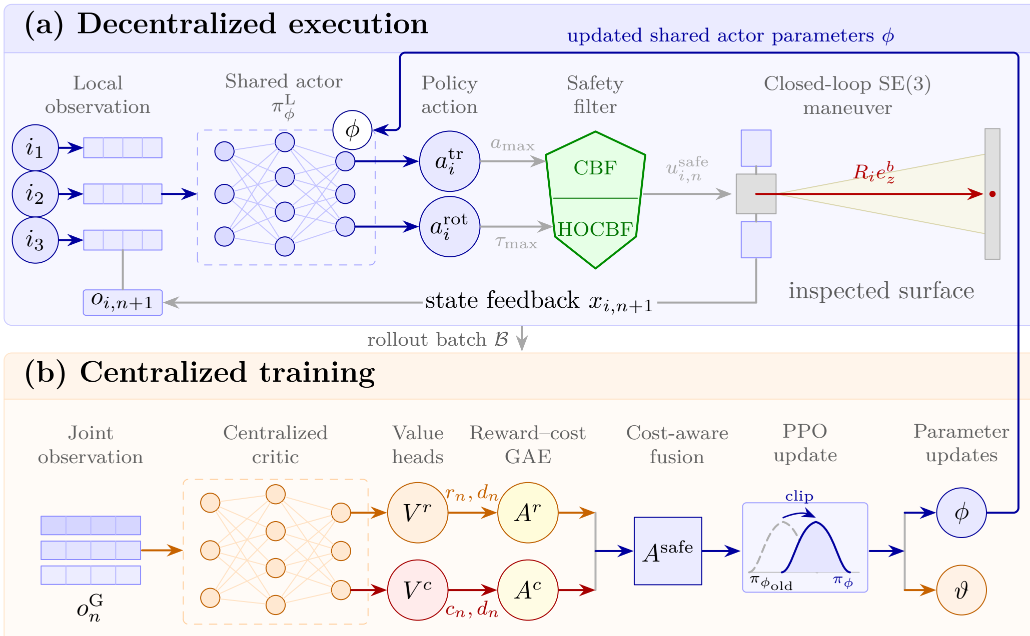

Figure 3: Safe-MAPPO low-level control architecture. During decentralized execution, each inspector evaluates the parameter-shared actor from its local goal-conditioned observation, and the resulting translational and rotational commands pass through the CBF/HOCBF safety projection before driving the closed-loop SE(3) dynamics. During centralized training, joint observations feed reward and safety-cost value heads, whose generalized advantage estimates are fused into a safety-aware advantage for the clipped PPO objective. Closed-loop rollout batches connect execution to training, while the updated shared actor parameters are returned to decentralized execution.

assigned-target event indicators are $I_{i,n+1}^{\mathrm{visit}} = \mathbb{I}\{z_j(t_n) = 0 \ \wedge \ z_j(t_{n+1}) = 1\}$ and $I_{i,n+1}^{\mathrm{near}} = \mathbb{I}\{z_j(t_n) = 0 \ \wedge \ d_{i,n+1}^g \leq r_{\mathrm{vis}}\}$. The near-viewpoint pointing reward is

$$\begin{aligned} r_{i,n}^{\mathrm{P}} = I_{i,n+1}^{\mathrm{near}} \left(w_{\mathrm{P}}[\chi_{ij,n+1} - a_{\mathrm{vis}}]_+ - w_{\mathrm{E}}[a_{\mathrm{vis}} - \chi_{ij,n+1}]_+ \right) \\ + I_{i,n+1}^{\mathrm{near}} \mathbb{I}\{\chi_{ij,n+1} \geq a_{\mathrm{vis}}\} \left(w_{\mathrm{sv}} e^{-\|v_{i,n+1}\|_2^2} + w_{\mathrm{s}\omega} e^{-\|\omega_{i,n+1}\|_2^2} \right), \end{aligned} \tag{26}$$

where $[x]_+ = \max(x, 0)$ and $\chi_{ij,n+1} = b_i(t_{n+1})^\top d_{ij}(t_{n+1})$. The complete per-inspector reward is written compactly as

$$\begin{aligned} r_{i,n} = w_{\mathrm{d}} \Delta d_{i,n}^g + w_\Phi \Delta \Phi_n + w_{\mathrm{new}} N_{n+1}^{\mathrm{new}} + w_{\mathrm{asg}} I_{i,n+1}^{\mathrm{visit}} + w_{\mathrm{goal}} I_{i,n+1}^{\mathrm{near}} + r_{i,n}^{\mathrm{P}} \\ - w_{\mathrm{tr}} \|\mathbf{a}_{i,n}^{\mathrm{tr}}\|_2^2 - w_{\mathrm{rot}} \|\mathbf{a}_{i,n}^{\mathrm{rot}}\|_2^2 - w_{\mathrm{v}} \|v_{i,n+1}\|_2^2 - w_\omega \|\omega_{i,n+1}\|_2^2 - w_0 \\ - \mathbf{w}_\epsilon^\top \boldsymbol{\epsilon}_{n+1} - w_{\Delta\mathrm{tr}} \Delta_{i,n}^{\mathrm{tr}} - w_{\Delta\mathrm{rot}} \Delta_{i,n}^{\mathrm{rot}}, \end{aligned} \tag{27}$$

Let $d^{\mathrm{I}}_{\min,n} = \min_{i<\ell} \|p_i(t_n) - p_\ell(t_n)\|_2$ and $c^{\mathrm{S}}_{\min,n} = \min_i c^{\mathrm{S}}_i(t_n)$ denote the minimum pairwise distance and station clearance at control instant n. The shaping-violation vector is

$$\boldsymbol{\epsilon}_{n+1} = \begin{bmatrix} [d_{\mathrm{safe}} + b_{\mathrm{I}} - d^{\mathrm{I}}_{\min,n+1}]_+ \\ [b_{\mathrm{S}} - c^{\mathrm{S}}_{\min,n+1}]_+ \\ [d_{\mathrm{safe}} + m_{\mathrm{I}} - d^{\mathrm{I}}_{\min,n+1}]_+ \\ [m_{\mathrm{S}} - c^{\mathrm{S}}_{\min,n+1}]_+ \end{bmatrix},$$

where $b_{\mathrm{I}}, b_{\mathrm{S}} \geq 0$ are boundary buffers and $m_{\mathrm{I}} \geq b_{\mathrm{I}}$, $m_{\mathrm{S}} \geq b_{\mathrm{S}}$ are approach margins. The execution-layer corrections are $\Delta^{\mathrm{tr}}_{i,n} = [\|a^{\mathrm{S}}_{i,n,\mathrm{safe}} - \bar{a}^{\mathrm{S}}_{i,n}\|_2 - \varepsilon_0]_+$ and $\Delta^{\mathrm{rot}}_{i,n} = [\|\tau^b_{i,n,\mathrm{safe}} - \bar{\tau}^b_{i,n}\|_2 - \varepsilon_0]_+$, where $a^{\mathrm{S}}_{i,n,\mathrm{safe}}$ and $\tau^b_{i,n,\mathrm{safe}}$ are the executed commands and $\varepsilon_0 > 0$ suppresses numerical noise. The translational quantity isolates the CBF projection, whereas the rotational quantity includes both the deterministic pointing guidance and the subsequent HOCBF projection relative to the actor torque; the pure HOCBF intervention is evaluated separately in Eq. (45). All scalar weighting coefficients in Eqs. (26) and (27) are nonnegative. In Eq. (27), w_0 is the constant per-step penalty and $\mathbf{w}_\epsilon \in \mathbb{R}^4_{\geq 0}$ weights the four shaping violations. The numerical settings are reported in Section 4.1.

Safe-MAPPO forms a separate team-level safety cost from pairwise proximity, station clearance, and translational-filter intervention. Let $d^{\mathrm{I}}_{\mathrm{c}} > 0$ and $c^{\mathrm{S}}_{\mathrm{c}} > 0$ denote the selected pairwise-distance and station-clearance cost margins, respectively. The safety cost broadcast to each inspector is

$$c_{i,n} = \lambda_{\mathrm{I}} \frac{[d^{\mathrm{I}}_{\mathrm{c}} - d^{\mathrm{I}}_{\min,n}]_+}{d^{\mathrm{I}}_{\mathrm{c}}} + \lambda_{\mathrm{S}} \frac{[c^{\mathrm{S}}_{\mathrm{c}} - c^{\mathrm{S}}_{\min,n}]_+}{c^{\mathrm{S}}_{\mathrm{c}}} + \lambda_\Delta \frac{1}{N} \sum_{\ell=1}^{N} \Delta^{\mathrm{tr}}_{\ell,n}. \tag{28}$$

The nonnegative coefficients λ_{I}, λ_{S}, and λ_Δ weight pairwise proximity, station proximity, and mean translational-filter intervention, respectively. This auxiliary cost is learned separately from the task reward and does not replace the pointwise CBF/HOCBF action projection.

Reward and cost generalized advantage estimates $A^r_{i,n}$ and $A^c_{i,n}$ and their value targets are computed independently with common discount and trace parameters (γ, λ). For rollout-batch mean μ_X and standard deviation σ_X, the actor uses

$$\begin{aligned} \widetilde{A}^{\mathrm{safe}}_{i,n} &= \frac{A^r_{i,n} - \mu_{A^r}}{\sigma_{A^r} + \varepsilon} - \beta_{\mathrm{c}} \frac{[A^c_{i,n}]_+}{\sigma_{A^c} + \varepsilon}, \\ A^{\mathrm{safe}}_{i,n} &= \frac{\widetilde{A}^{\mathrm{safe}}_{i,n} - \mu_{\widetilde{A}^{\mathrm{safe}}}}{\sigma_{\widetilde{A}^{\mathrm{safe}}} + \varepsilon}, \end{aligned} \tag{29}$$

where $\varepsilon > 0$ stabilizes normalization and $\beta_c \geq 0$ controls safety shaping. With likelihood ratio $\varrho_{i,n} = \pi^{\mathrm{L}}_{\phi}(\mathbf{a}_{i,n} \mid o_{i,n})/\pi^{\mathrm{L}}_{\phi_{\mathrm{old}}}(\mathbf{a}_{i,n} \mid o_{i,n})$, the complete loss is

$$\begin{aligned}\mathcal{L}_{\text{Safe-MAPPO}} = & -\mathbb{E}\Big[\min\Big(\varrho_{i,n}A^{\mathrm{safe}}_{i,n}, \operatorname{clip}(\varrho_{i,n}, 1-\epsilon_\pi, 1+\epsilon_\pi)A^{\mathrm{safe}}_{i,n}\Big)\Big] \\ & + c^r_V\mathcal{L}^r_V + c^c_V\mathcal{L}^c_V - c_{\mathrm{H}}\mathbb{E}[\mathcal{H}(\pi^{\mathrm{L}}_{\phi})],\end{aligned} \tag{30}$$

where $\mathcal{L}^r_V$ and $\mathcal{L}^c_V$ are PPO-clipped value losses and $\mathcal{H}$ is policy entropy. The safety-cost critic favors trajectories requiring less intervention, while the following CBF/HOCBF layer projects the executed control input.

3.4. CBF/HOCBF Safety Filter for SE(3) *Execution*

The execution layer applies two control projections after the low-level policy and before each dynamics step: a translational projection for collision avoidance and a rotational projection for surface pointing. Both projections use relative-degree-two barrier conditions and elastic QPs based on the standard CBF-QP and HOCBF constructions [20, 21]. The low-level policy is evaluated every Δt_π, while each nominal policy command is held for Q safety substeps of duration $\Delta t_{\mathrm{s}} = \Delta t_\pi/Q$ and both QPs are recomputed from the propagated state at every substep. This dual-rate execution limits the interval over which a projected input is held without a new barrier evaluation, addressing the inter-sample effect that arises when continuous-time CBF conditions are implemented in sampled-data systems [22].

For stacked state $x^{\mathrm{tr}} = \operatorname{col}(p_1, \ldots, p_N, v_1, \ldots, v_N)$ and acceleration $\mathbf{a}^{\mathrm{S}} = \operatorname{col}(a^{\mathrm{S}}_1, \ldots, a^{\mathrm{S}}_N)$, the translational subsystem is the control-affine double integrator $\dot{x}^{\mathrm{tr}} = f_{\mathrm{tr}}(x^{\mathrm{tr}}) + g_{\mathrm{tr}}(x^{\mathrm{tr}})\mathbf{a}^{\mathrm{S}}$. Let $\mathcal{K}_{\mathrm{tr}}$ index all pairwise barriers in Eq. (12) and all inflated station-geometry barriers in Eq. (11), and denote an indexed barrier by h^{tr}_r for $r \in \mathcal{K}_{\mathrm{tr}}$. Using this inflated-clearance notation, the QP applies sampling-compensated control barriers

$$\widetilde{h}^{\mathrm{S}}_{i\ell} = d^{\mathrm{S}}_{\ell}(p_i; m^{\mathrm{S}}_{\mathrm{s}}), \qquad \widetilde{h}^{\mathrm{I}}_{i\ell} = \|p_i - p_\ell\|^2_2 - \left(d_{\mathrm{safe}} + m^{\mathrm{I}}_{\mathrm{s}}\right)^2, \tag{31}$$

where $m^{\mathrm{S}}_{\mathrm{s}} = v_{\max}\Delta t_{\mathrm{s}}$ and $m^{\mathrm{I}}_{\mathrm{s}} = 2v_{\max}\Delta t_{\mathrm{s}}$ provide first-order one- and two-inspector travel allowances associated with the stored speed bound over one safety update interval. The nominal analytic safety metrics continue to use the bounding-body barriers in Eqs. (11) and (12); the additional margins in Eq. (31) shrink only the control set used by the QP. Each position barrier has relative degree two with respect to acceleration, giving

$$\begin{aligned}\psi_{r,1}^{\mathrm{tr}} &= L_{f_{\mathrm{tr}}}\widetilde{h}_r^{\mathrm{tr}} + k_1^{\mathrm{tr}}\widetilde{h}_r^{\mathrm{tr}},\\ L_{f_{\mathrm{tr}}}\psi_{r,1}^{\mathrm{tr}} + L_{g_{\mathrm{tr}}}\psi_{r,1}^{\mathrm{tr}}\mathbf{a}_n^{\mathrm{S}} + k_2^{\mathrm{tr}}\psi_{r,1}^{\mathrm{tr}} &\geq -\delta_{r,n}^{\mathrm{tr}}, \qquad k_1^{\mathrm{tr}}, k_2^{\mathrm{tr}} > 0.\end{aligned} \tag{32}$$

Let $\bar{\mathbf{a}}_n^{\mathrm{S}} = \mathrm{col}(\bar{a}_{1,n}^{\mathrm{S}}, \ldots, \bar{a}_{N,n}^{\mathrm{S}})$ be the nominal policy command defined in Eq. (23). The translational safety projection is the elastic QP

$$\begin{aligned}\left(\mathbf{a}_{n,\mathrm{safe}}^{\mathrm{S}}, \boldsymbol{\delta}_n^{\mathrm{tr}}, \boldsymbol{\xi}_n^{\mathrm{tr},+}, \boldsymbol{\xi}_n^{\mathrm{tr},-}\right) = \underset{\mathbf{a}^{\mathrm{S}},\boldsymbol{\delta},\boldsymbol{\xi}^+,\boldsymbol{\xi}^-}{\arg\min} \quad & \left\|\mathbf{a}^{\mathrm{S}} - \bar{\mathbf{a}}_n^{\mathrm{S}}\right\|_2^2 + \rho_h^{\mathrm{tr}}\mathbf{1}^\top\boldsymbol{\delta} + \rho_u^{\mathrm{tr}}\mathbf{1}^\top\left(\boldsymbol{\xi}^+ + \boldsymbol{\xi}^-\right)\\ \text{subject to} \quad & L_{f_{\mathrm{tr}}}\psi_{r,1}^{\mathrm{tr}} + L_{g_{\mathrm{tr}}}\psi_{r,1}^{\mathrm{tr}}\mathbf{a}^{\mathrm{S}} + k_2^{\mathrm{tr}}\psi_{r,1}^{\mathrm{tr}} \geq -\delta_r, \quad r \in \mathcal{K}_{\mathrm{tr}},\\ & -a_{\max}\mathbf{1} - \boldsymbol{\xi}^- \preceq \mathbf{a}^{\mathrm{S}} \preceq a_{\max}\mathbf{1} + \boldsymbol{\xi}^+,\\ & \boldsymbol{\delta} \succeq \mathbf{0}, \quad \boldsymbol{\xi}^+ \succeq \mathbf{0}, \quad \boldsymbol{\xi}^- \succeq \mathbf{0}.\end{aligned} \tag{33}$$

The station capsule and OBBs are inflated by the attitude-independent inspector bounding radius, while d_{safe} includes the bounding radii of both inspectors, so the center-position barriers conservatively protect the finite-size rigid bodies without introducing attitude into the collision QP.

The rotational projection is activated according to a configurable proximity rule around the currently assigned viewpoint. For $j = \sigma_i^k$, define the activation indicator and the guided nominal torque as

$$\begin{aligned}\mu_{i,n}^{\mathrm{P}} &= \mathbb{I}\left\{\left\|p_{i,n} - p_j^v\right\|_2 < r_{\mathrm{P}}\right\},\\ \bar{\tau}_{i,n}^{\mathrm{P}} &= \mathrm{sat}_{\tau_{\max}}\left(\bar{\tau}_{i,n}^b + \mu_{i,n}^{\mathrm{P}}\left[k_{\mathrm{p}}R_{i,n}^\top\left(b_{i,n} \times d_{ij,n}\right) - k_{\mathrm{d}}\omega_{i,n}\right]\right),\end{aligned} \tag{34}$$

where r_{P} is the activation radius, $k_{\mathrm{p}}, k_{\mathrm{d}} > 0$ are feedback gains, and $r_{\mathrm{P}} = \infty$ gives the continuous surface-pointing mode used here. The current $d_{ij,n}$ is held fixed while evaluating rotational Lie derivatives, and the relative-degree-two auxiliary barrier is $\psi_{i,1}^{\mathrm{P}} = L_{f_{\mathrm{rot}}}h_{ij}^{\mathrm{P}} + k_1^{\mathrm{P}}h_{ij}^{\mathrm{P}}$. Let $\mathcal{N}_n^{\mathrm{P}} = \{i \in \mathcal{N} \mid \mu_{i,n}^{\mathrm{P}} = 1\}$ denote the active pointing set and let $\bar{\boldsymbol{\tau}}_n^{\mathrm{P}} = \mathrm{col}(\bar{\tau}_{1,n}^{\mathrm{P}}, \ldots, \bar{\tau}_{N,n}^{\mathrm{P}})$. The rotational projection is

$$\begin{aligned}\left(\boldsymbol{\tau}_{n,\mathrm{safe}}^b, \boldsymbol{\delta}_n^{\mathrm{P}}, \boldsymbol{\xi}_n^{\mathrm{P},+}, \boldsymbol{\xi}_n^{\mathrm{P},-}\right) = \underset{\boldsymbol{\tau}^b,\boldsymbol{\delta},\boldsymbol{\xi}^+,\boldsymbol{\xi}^-}{\arg\min} \quad & \left\|\boldsymbol{\tau}^b - \bar{\boldsymbol{\tau}}_n^{\mathrm{P}}\right\|_2^2 + \rho_h^{\mathrm{P}}\mathbf{1}^\top\boldsymbol{\delta} + \rho_u^{\mathrm{P}}\mathbf{1}^\top\left(\boldsymbol{\xi}^+ + \boldsymbol{\xi}^-\right)\\ \text{subject to} \quad & L_{f_{\mathrm{rot}}}\psi_{i,1}^{\mathrm{P}} + L_{g_{\mathrm{rot}}}\psi_{i,1}^{\mathrm{P}}\boldsymbol{\tau}^b + k_2^{\mathrm{P}}\psi_{i,1}^{\mathrm{P}} \geq -\delta_i, \quad i \in \mathcal{N}_n^{\mathrm{P}},\\ & -\tau_{\max}\mathbf{1} - \boldsymbol{\xi}^- \preceq \boldsymbol{\tau}^b \preceq \tau_{\max}\mathbf{1} + \boldsymbol{\xi}^+,\\ & \boldsymbol{\delta} \succeq \mathbf{0}, \quad \boldsymbol{\xi}^+ \succeq \mathbf{0}, \quad \boldsymbol{\xi}^- \succeq \mathbf{0}, \qquad k_2^{\mathrm{P}} > 0.\end{aligned} \tag{35}$$

For an inactive inspector, the bounded nominal torque is passed through unchanged. The nonnegative coefficients ρ_h^{tr}, ρ_u^{tr}, ρ_h^{P}, and ρ_u^{P} penalize barrier and actuator-bound relaxation, with large values used to prioritize constraint satisfaction over nominal-command tracking. When all relaxation variables vanish and the initial state satisfies both the original barrier and first auxiliary-barrier conditions, Eqs. (32) and (35) recover the standard relative-degree-two forward-invariance conditions; the elastic variables preserve numerical feasibility when saturated commands or competing constraints make the unrelaxed QP infeasible. When the elastic variables vanish, the speed-based margins in Eq. (31) tighten the admissible control set against first-order inter-sample travel. Because the implemented semi-implicit update applies acceleration before clipping the stored velocity and because nonzero relaxation can admit constraint residuals, these margins are treated as sampled-data compensation rather than a standalone invariance theorem; nominal analytic margins, control margins, QP relaxation, and detailed MuJoCo contacts are therefore all reported.

The safe input remains $u_{i,n,\mathrm{safe}} = (a^{\mathrm{S}}_{i,n,\mathrm{safe}}, \tau^b_{i,n,\mathrm{safe}})$. The corresponding body-frame force $f^b_{i,n,\mathrm{safe}} = m_i R_{i,n}^\top a^{\mathrm{S}}_{i,n,\mathrm{safe}}$ is used only for rigid-body integration. CBFpy evaluates the Lie derivatives and assembles the safety-filter QPs in JAX [34]. QPax solves the resulting batched elastic programs with a primal–dual interior-point method [35]. This implementation keeps the safety projections compatible with vectorized training and closed-loop evaluation. The numerical barrier gains, relaxation penalties, actuator bounds, and integration interval are reported in Section 4.1.

3.5. Staged Training and Closed-Loop Execution

The hierarchical policy is optimized in two stages to prevent the high-level assignment distribution and low-level motion policy from changing simultaneously. In the first stage, the low-level environment samples nonduplicate targets uniformly from the open-target set and refreshes the assignments after H_{goal} control instants or when an assigned target is completed. This random-goal curriculum exposes the shared actor to the complete viewpoint set instead of coupling it to a nearest-target rule. The actor and two centralized critic heads are optimized using the Safe-MAPPO loss in Eq. (30), while the CBF/HOCBF filters remain active within every rollout. After convergence, the low-level parameters are fixed at $(\phi^\star, \vartheta^\star)$.

The second stage initializes the GNN allocator randomly and trains it directly at the option time scale through closed-loop PPO with the frozen

Algorithm 1 Staged training and safe hierarchical execution

Input: $\mathcal{E}_{\mathrm{SE}(3)}$, H_{goal}, H_{opt}, H_{hold}, $K_{\max}$, Q, U_{L}, U_{H}, and U_{val}.

Stage I: Goal-conditioned Safe-MAPPO

$(\phi, \vartheta) \leftarrow$ Initialize().
for $u = 1, \ldots, U_{\mathrm{L}}$ **do**
  $\boldsymbol{\sigma} \sim$ UnifWOR($\mathcal{M}_{\mathrm{open}}$) every H_{goal} steps or after target completion.
  $\mathcal{B}_{\mathrm{L}} \leftarrow \mathrm{Rollout}(\mathcal{E}_{\mathrm{SE}(3)}, \pi^{\mathrm{L}}_{\phi}, \mathcal{S}_{\mathrm{CBF}}; \boldsymbol{\sigma})$.
  $(A^r, A^c, R^r, R^c) \leftarrow \mathrm{GAE}(\mathcal{B}_{\mathrm{L}})$; update (ϕ, ϑ) using $\mathcal{L}_{\text{Safe-MAPPO}}$.
end for; $(\phi^\star, \vartheta^\star) \leftarrow \mathrm{stopgrad}(\phi, \vartheta)$.

Stage II: Graph-based option policy

$(\theta, \eta) \leftarrow$ Initialize(); $J^\star \leftarrow -\infty$.
for $u = 1, \ldots, U_{\mathrm{H}}$ **do**
  $\mathcal{B}_{\mathrm{H}} \leftarrow \mathrm{Rollout}(\mathcal{E}_{\mathrm{SE}(3)}, \pi^{\mathrm{H}}_{\theta}, \pi^{\mathrm{L}}_{\phi^\star}, \mathcal{S}_{\mathrm{CBF}})$.
  Compute $(\widetilde{G}^{\mathrm{H}}_k, \widehat{A}^{\mathrm{H}}_k)$ and update (θ, η) using $\mathcal{L}_{\mathrm{H}}(\theta, \eta)$.
  Every U_{val} updates, if $J^{\mathrm{val}}_u > J^\star$ then $(\theta^\star, J^\star) \leftarrow (\theta, J^{\mathrm{val}}_u)$.
end for; $\theta^\star \leftarrow \mathrm{stopgrad}(\theta^\star)$.

Stage III: Closed-loop execution

$(x_0, \mathbf{z}_0) \leftarrow \mathrm{Reset}(\mathcal{E}_{\mathrm{SE}(3)})$; $n \leftarrow 0$.
while $C_n < 1$ and $n < n_{\max}$ **do**
  if $n \bmod H_{\mathrm{opt}} = 0$ **then** $\widehat{S}^k \leftarrow \mathrm{GNN}_{\theta^\star}(G^k)$; $\boldsymbol{\sigma}^k \sim \mathrm{SampleWOR}(\widehat{S}^k, \mathbf{z}_n, H_{\mathrm{hold}})$; **end if**.
  $\mathbf{a}_n \leftarrow \tanh(\mu_{\phi^\star}(o_n, \boldsymbol{\sigma}^k))$; $(\bar{\mathbf{a}}^{\mathrm{S}}_n, \bar{\boldsymbol{\tau}}^b_n) \leftarrow \mathrm{Scale}(\mathbf{a}_n)$.
  for $q = 0, \ldots, Q-1$ **do**
    Apply $\mathrm{QP}_{\mathrm{tr}}$ and QP_{P}, then propagate $x_{n,q+1} \leftarrow \mathcal{F}^{\Delta t_{\mathrm{s}}}_{\mathrm{SE}(3)}(x_{n,q}, u_{n,q,\mathrm{safe}})$.
  end for; $x_{n+1} \leftarrow x_{n,Q}$; $\mathbf{z}_{n+1} \leftarrow \mathrm{Visit}(x_{n+1}, \mathbf{z}_n)$.
  $n \leftarrow n + 1$; **end while**.

Output: $(\theta^\star, \phi^\star, \vartheta^\star)$ and $\{x_n, \boldsymbol{\sigma}^k\}_{n=0}^{n_f}$, where $n_f \le n_{\max}$.

Safe-MAPPO controller and both safety filters active. An option contains H_{opt} low-level control instants, and each high-level episode contains at most $K_{\max}$ options. Unique assignments are sampled without replacement from the learned edge scores.

Let $\mathcal{F}^k$ contain the unfinished assignment edges whose minimum commitment interval has not elapsed, let $\mathcal{I}^k_0$ and $\mathcal{J}^k_0$ contain their inspector and target endpoints, respectively, and let $N^{\mathrm{free}}_k = N - |\mathcal{F}^k|$ denote the number of inspectors requiring a new decision. The number of newly sampled edges is $R_k = \min\{N^{\mathrm{free}}_k, |\mathcal{M}^k_{\mathrm{open}} \setminus \mathcal{J}^k_0|\}$, which explicitly allows some inspectors to remain unassigned when fewer open targets are available. For these decisions, let $e^k_r = (i_r, j_r)$ denote the rth sampled edge, augment $\mathcal{I}^k_{r-1}$

and $\mathcal{J}_{r-1}^k$ with the endpoints of the previously sampled edges, and define $\mathcal{E}_r^k = (\mathcal{N} \setminus \mathcal{I}_{r-1}^k) \times (\mathcal{M}_{\text{open}}^k \setminus \mathcal{J}_{r-1}^k)$. The probability of the ordered edge-selection trace is

$$\pi_\theta^{\mathrm{H}}(e_{1:R_k}^k \mid G^k, \mathcal{F}^k) = \prod_{r=1}^{R_k} \frac{\exp\left(\hat{s}_{i_r j_r}^k\right)}{\sum_{(i,j)\in\mathcal{E}_r^k} \exp\left(\hat{s}_{ij}^k\right)}, \tag{36}$$

where the retained edges in $\mathcal{F}^k$ are inserted directly into σ^k and the sampled trace supplies the R_k new assignments $\sigma_{i_r}^k = j_r$. An inspector that is not selected because $R_k < N_k^{\text{free}}$ retains its previous completed goal as a pose-hold reference, or receives $\varnothing$ and holds its current pose if no previous goal exists. Retained edges are excluded from the policy log probability and entropy because they are option-state constraints rather than new actions selected at update k. No behavior cloning, nearest-target labels, additive distance prior, or surrogate transition model is used in this stage, so every allocator update is generated by the same rigid-body and safety-filtered closed loop used for evaluation. For the coverage task, the closed-loop option reward is

$$\begin{aligned} r_k^{\mathrm{H}} = {} & \alpha_{\text{new}} N_{k+1}^{\mathrm{H,new}} + \alpha_C \left(C_{k+1} - C_k\right) \\ & - \alpha_D \bar{D}_k^{\mathrm{H}} - \alpha_U \bar{U}_k^{\mathrm{H}} - \alpha_B B_{k+1}^{\mathrm{H}} - \alpha_S S_k^{\mathrm{H}} - \alpha_\Delta \bar{\Delta}_{a,k} - \alpha_0, \end{aligned} \tag{37}$$

where $N_{k+1}^{\mathrm{H,new}}$ is the number of newly completed targets, $\bar{D}_k^{\mathrm{H}}$ is the team distance traveled, $\bar{U}_k^{\mathrm{H}}$ is the accumulated low-level control effort, B_{k+1}^{H} is the workload imbalance, S_k^{H} counts switches away from unfinished assignments, and $\bar{\Delta}_{a,k}$ is the mean translational-filter intervention during option k. Let K denote the number of high-level options actually executed before complete coverage or the episode horizon is reached. The reward at the last active option $k = K-1$ receives the terminal increment $\alpha_{\text{succ}}\mathbb{I}\{C_K = 1\} + \alpha_{\text{term}} C_K$, and all coefficients in Eq. (37) are nonnegative. The reward contains no positive term proportional to traveled distance or reduction in assigned-goal distance; consequently, extending an otherwise equivalent route can only decrease the return through $\bar{D}_k^{\mathrm{H}}$, $\bar{U}_k^{\mathrm{H}}$, and α_0. The discounted return $G_k^{\mathrm{H}} = \sum_{l=k}^{K-1} \gamma_{\mathrm{H}}^{l-k} r_l^{\mathrm{H}}$ is scaled as $\widetilde{G}_k^{\mathrm{H}} = G_k^{\mathrm{H}}/s_V$ and combined with centralized value network V_η^{H} to form the standardized option advantage $\widehat{A}_k^{\mathrm{H}}$. Only active options contribute to return and value statistics, and actor terms exclude persistence-only decisions and post-terminal padding. Let $\varrho_k^{\mathrm{H}}(\theta) = \pi_\theta^{\mathrm{H}}(e_{1:R_k}^k \mid G^k, \mathcal{F}^k)/\pi_{\theta_{\text{old}}}^{\mathrm{H}}(e_{1:R_k}^k \mid G^k, \mathcal{F}^k)$ denote the exact likelihood ratio of the ordered sample-without-replacement trace over newly selected edges. The allocator and value network minimize the clipped option-level PPO objective

$$\begin{aligned}\mathcal{L}_{\mathrm{H}}(\theta,\eta) = & -\mathbb{E}\Big[\min\Big(\varrho_k^{\mathrm{H}}(\theta)\widehat{A}_k^{\mathrm{H}}, \mathrm{clip}\Big(\varrho_k^{\mathrm{H}}(\theta), 1-\epsilon_{\mathrm{H}}, 1+\epsilon_{\mathrm{H}}\Big)\,\widehat{A}_k^{\mathrm{H}}\Big)\Big] \\ & + c_V^{\mathrm{H}}\mathbb{E}\Big[\Big(V_\eta^{\mathrm{H}}(G^k) - \widetilde{G}_k^{\mathrm{H}}\Big)^2\Big] - c_{\mathrm{ent}}^{\mathrm{H}}\mathbb{E}\Big[\mathcal{H}(\pi_\theta^{\mathrm{H}})\Big]\,,\end{aligned} \tag{38}$$

where ϵ_{H} is the high-level clipping threshold and c_V^{H} and $c_{\mathrm{ent}}^{\mathrm{H}}$ are the value and entropy coefficients. Checkpoint selection uses a separate fixed validation batch and does not access the final test missions.

During execution, the GNN samples Eq. (36) every H_{opt} control steps, while unfinished assignments persist for at least H_{hold} options. Between assignment events, newly completed goals remain fixed until the next boundary, and the deterministic mean action of the frozen low-level actor is processed by the two safety QPs and then applied to the rigid-body dynamics. Execution terminates when all targets have been inspected or the episode horizon is reached. Algorithm 1 summarizes the low-level training, high-level training, and closed-loop execution stages. Here, $Q = \Delta t_\pi/\Delta t_{\mathrm{s}}$, U_{L} and U_{H} are the low- and high-level update budgets, U_{val} is the validation interval, and UnifWOR and SampleWOR denote uniform and policy-weighted sampling without replacement, respectively. The operator stopgrad freezes each selected network.

4. Experimental Evaluation

The framework is evaluated using mission efficiency, pairwise separation, Gateway clearance, and pointing accuracy. All methods are evaluated under a common SE(3) simulation setting and matched initial-condition batches to ensure consistent comparisons. The evaluation first reports the training behavior and closed-loop performance of the complete framework, followed by controlled comparisons and ablations of the graph-based task allocator, low-level Safe-MAPPO controller, and CBF/HOCBF safety filters. The experiments address three questions: whether the complete hierarchy converges to reliable inspection behavior, whether graph context improves learned assignment relative to graph-free policies, and how the safety critic and execution filters separately affect mission and constraint metrics.

4.1. Experimental Setup and Evaluation Metrics

The inspected structure is instantiated as the Lunar Gateway space station, whose modular architecture provides a representative station-scale target for close-proximity inspection [1]. Figure 4 presents a high-resolution frame from the same MuJoCo replay used for the animated inspection visualization, preserving the recorded spacecraft states, waypoint markers, sensor cones, and recent paths. All compared methods use the same Gateway-centered scenario and identical physical parameters. The vectorized training and quantitative evaluation rollouts propagate the rigid-body dynamics, policies, and safety QPs in JAX [36], whereas the resulting state trajectories are replayed with the corresponding inspector and station models in MuJoCo [37] for rendering and contact validation. The MuJoCo replay does not modify the evaluated trajectory and therefore provides a separate collision-geometry check against the detailed simulation assets.

The nominal scenario contains three inspector spacecraft and eight inspection targets selected from the station inspection set. Each target comprises a desired viewpoint and its corresponding station surface point. The nominal initial inspector positions in $\mathcal{F}_{\mathrm{S}}$ are

$$\bar{P}_0 = \begin{bmatrix} -8.0 & -16.0 & 0.5 \\ 3.5 & 20.0 & 1.0 \\ 20.0 & -10.0 & 0.0 \end{bmatrix} \text{ m.} \tag{39}$$

At each reset, the position and translational velocity of every inspector are perturbed independently by zero-mean Gaussian noise with standard deviations of 2.0 m and $0.02\,\mathrm{m\,s^{-1}}$, respectively. A perturbed inspector position with less than 0.25 m clearance from the sampled-data station control envelope is replaced by its nominal start, and the complete position set is replaced by the nominal starts if the resulting team violates the sampled-data pair-separation threshold. The initial attitude is sampled by normalizing a quaternion of the form $[1, \epsilon_q^\top]^\top$ with $\epsilon_q \sim \mathcal{N}(0, 0.02^2 I_3)$ and converting it to $R_i(0) \in \mathrm{SO}(3)$, while the initial body angular velocity is sampled from $\mathcal{N}(0, 0.01^2 I_3)\,\mathrm{rad\,s^{-1}}$. An episode terminates when all eight targets have been inspected or when the maximum horizon of 2400 integration steps is reached.

Table 1 lists the principal control, geometry, and safety-filter parameters. Each inspector uses the same mass and diagonal inertia matrix, and its normalized six-dimensional policy action is mapped to a bounded station-frame acceleration and body-frame torque before safety filtering. The acceleration and torque limits in Table 1 therefore bound the nominal commands

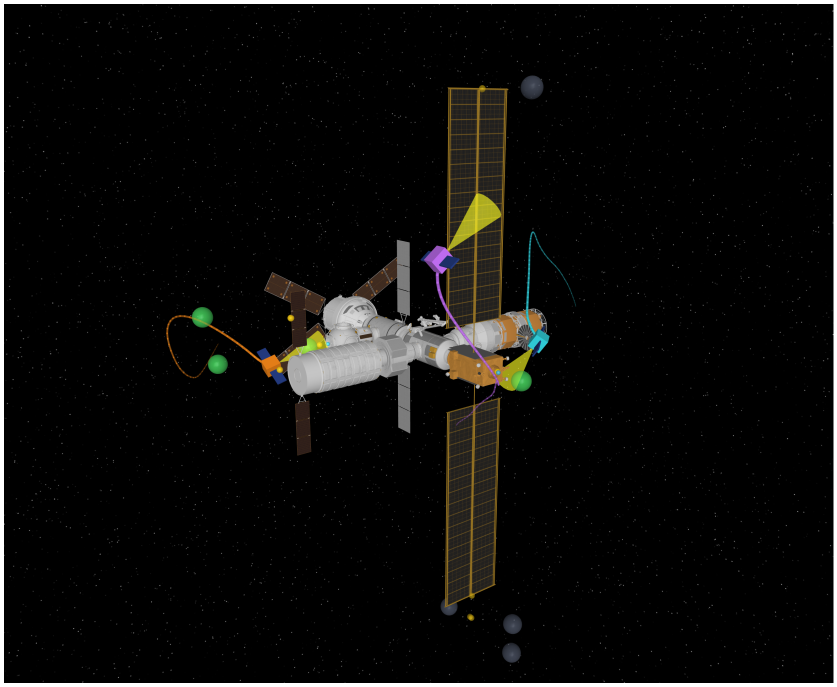

Figure 4: High-resolution oblique MuJoCo frame from a successful rollout of the complete hierarchical method at 50% coverage. The orange, cyan, and purple inspector spacecraft are assigned to distinct module, lower-solar-array, and upper-solar-array viewpoints, respectively. Their recent paths use the same colors; all eight inspection viewpoints are marked by spheres, with green and gray denoting visited and unvisited viewpoints, respectively, while the yellow cones show the instantaneous sensor-axis directions.

and define relaxable QP constraints; any executed-input excess is exposed separately through the reported control-relaxation diagnostics. The station keep-out model contains one axial capsule and 26 OBB primitives extracted from the MuJoCo collision geometry. For each compiled collision mesh, its world-frame geom center and orientation define the OBB frame, while the componentwise maximum absolute local vertex coordinates define half extents that conservatively enclose the complete mesh. The 2.3 m inspector sphere encloses the detailed bus and both solar-panel collision geoms, and its radius inflates the capsule and every OBB half extent, while the minimum center separation includes the bounding radii of two inspectors and a numerical clearance margin. The experimental pointing activation radius is unbounded, so the guided pointing controller and attitude HOCBF continuously track the surface point associated with the active assignment. Within

Table 1: Principal control, geometry, and safety-filter parameters.

Parameter	Symbol	Value
Low-level policy interval	Δt_π	$0.25\,\mathrm{s}$
Safety-filter interval	Δt_s	$0.05\,\mathrm{s}$
Inspector mass	m	$12.0\,\mathrm{kg}$
Body inertia	J	$\mathrm{diag}(2.6, 11.2, 10.6)\,\mathrm{kg}\cdot\mathrm{m}^2$
Nominal acceleration limit	a_max	$0.35\,\mathrm{m\,s^{-2}}$
Nominal torque limit	τ_max	$0.45\,\mathrm{N\,m}$
Speed bound	v_max	$1.2\,\mathrm{m\,s^{-1}}$
Angular-speed bound	ω_max	$0.25\,\mathrm{rad\,s^{-1}}$
Inspector bounding radius	r_B	$2.3\,\mathrm{m}$
Minimum center separation	d_safe	$4.8\,\mathrm{m}$
Visit radius	r_vis	$4.0\,\mathrm{m}$
Minimum pointing alignment	a_vis	0.85
High-level assignment interval	H_opt	64 steps ($16\,\mathrm{s}$)
Minimum assignment hold	H_hold	3 options ($48\,\mathrm{s}$)
Translational HOCBF gains	$(k_1^\mathrm{tr}, k_2^\mathrm{tr})$	$(0.8, 1.2)$
Pointing HOCBF gains	$(k_1^\mathrm{P}, k_2^\mathrm{P})$	$(3.0, 4.5)$
Pointing feedback gains	$(k_\mathrm{p}, k_\mathrm{d})$	$(1.8, 2.3)$
Sample-hold station margin	m_s^S	$0.06\,\mathrm{m}$
Sample-hold pair margin	m_s^I	$0.12\,\mathrm{m}$
QP relaxation penalties	(ρ_h, ρ_u)	$(10^5, 10^5)$

every $0.05\,\mathrm{s}$ safety substep, the Lie-group backend advances attitude through the $\mathrm{SO}(3)$ exponential map and applies the stated translational- and angular-speed limits to the stored state after integration. The clearance feature in Eq. (15) uses $c_\mathrm{S} = 10\,\mathrm{m}$, and both elastic QPs use the termination tolerance $\varepsilon_\mathrm{QP} = 10^{-4}$.

Each evaluation episode samples eight active targets without replacement from a fixed catalog of 16 feasible Gateway viewpoints and perturbs the three inspector initial positions with independent zero-mean Gaussian noise of $2.0\,\mathrm{m}$ standard deviation under the collision-safe fallback rule described above. The low-level Safe-MAPPO policy is optimized under random nonduplicate target assignments that are refreshed every 128 steps or after an assigned target is inspected. Its curriculum first trains on four and then eight fixed viewpoints for 800 updates each, performs a 400-update station-clearance adaptation, expands the task to eight active targets sampled from the 16-viewpoint catalog for 600 updates, adapts the converged controller to Lie-group attitude integration for 100 updates, and concludes with a 30-update dual-rate safety adaptation followed by 30 updates with explicit completed-goal or pose-hold semantics under the surface-pointing gains reported in Table 1. The initial four-viewpoint phase uses a 1200-step episode horizon, and every subsequent low-level phase uses the final 2400-step horizon. Every phase uses 64 parallel environments, 64-step rollouts, four optimization epochs per update, eight minibatches, and multilayer perceptrons (MLPs) with 128 hidden units per layer. The complete low-level curriculum contains 2760 policy updates and 11,304,960 vectorized environment transitions, including every initialization and adaptation stage in the reported checkpoint lineage. The curriculum schedule, terminal low-level checkpoint, and formal task protocol were frozen before the multi-seed allocator study and were not modified using either independent test batch. The discount factor is $\gamma = 0.99$ and the generalized-advantage parameter is $\lambda = 0.95$ throughout, while the learning rate is annealed across the curriculum from 3×10^{-4} to 10^{-5}, the PPO clipping threshold from 0.2 to 0.1, and the entropy coefficient from 5×10^{-3} to zero. The safety-advantage coefficient β_c is 0.25 during the four- and eight-viewpoint pretraining phases and 0.02 after expansion to the 16-viewpoint catalog, while the cost-value coefficient remains 0.25. The final low-level adaptation uses the following reward and regularization weights:

$$\begin{aligned}
(w_{\mathrm{d}}, w_{\Phi}, w_{\mathrm{new}}, w_{\mathrm{asg}}, w_{\mathrm{goal}}) &= (2, 4, 50, 20, 1), \\
(w_{\mathrm{P}}, w_{\mathrm{E}}, w_{\mathrm{sv}}, w_{\mathrm{s}\omega}) &= (2, 0.5, 0.1, 0.1), \\
(w_{\mathrm{tr}}, w_{\mathrm{rot}}, w_{\mathrm{v}}, w_{\omega}, w_0) &= (0.008, 0.004, 0.05, 0.05, 0.05), \\
\mathbf{w}_{\epsilon} &= (8, 20, 0.30, 0.80)^{\top}, \\
(w_{\Delta\mathrm{tr}}, w_{\Delta\mathrm{rot}}) &= (0.04, 0.02), \qquad \varepsilon_0 = 10^{-4}.
\end{aligned} \tag{40}$$

The corresponding shaping buffers are $(b_{\mathrm{I}}, b_{\mathrm{S}}) = (0, 0)\,\mathrm{m}$ and $(m_{\mathrm{I}}, m_{\mathrm{S}}) = (0.62, 1.06)\,\mathrm{m}$, while the final safety cost uses $(d_{\mathrm{c}}^{\mathrm{I}}, c_{\mathrm{c}}^{\mathrm{S}}) = (5.42, 1.0)\,\mathrm{m}$ and $(\lambda_{\mathrm{I}}, \lambda_{\mathrm{S}}, \lambda_{\Delta}) = (1, 1, 0.05)$. Each high-level allocator is initialized randomly and optimized for 40 closed-loop PPO updates with 64 parallel missions per update, at most 38 option decisions per mission, and four optimization epochs per update. The learning rate is 3×10^{-4}, the discount factor is $\gamma_{\mathrm{H}} = 0.98$, the clipping threshold is $\epsilon_{\mathrm{H}} = 0.2$, and the entropy coefficient is 10^{-4}. The high-level reward coefficients are fixed before the multi-seed comparison and

Table 2: High-level coverage-task reward coefficients.

Coefficient	Reward component	Value
α_{new}	Newly inspected targets	33.0
α_C	Coverage increment	48.0
α_D	Team path length	0.20
α_U	Low-level control effort	0.003
α_B	Workload imbalance	0.20
α_S	Premature assignment switch	0.10
α_Δ	Translational-filter intervention	0.80
α_0	Active-option duration	1.0
$(\alpha_{\text{succ}}, \alpha_{\text{term}})$	Terminal success and coverage	$(80.0, 70.0)$

are listed in Table 2.

Assignments are updated only at fixed 64-step option boundaries, an unfinished assignment is retained for at least three options, and both rules are applied identically to every learned and prescribed allocator. If fewer open targets than inspectors remain, each unselected inspector retains its previous completed goal as a pose-hold reference or holds its current pose when no previous goal is available. Before training and after every five updates, the current stochastic assignment policy is evaluated on a fixed set of 128 validation missions generated with seed 2026 and a fixed assignment random stream, and the checkpoint with the largest validation score is selected without reference to the final test batches. Three independent random initializations and rollout seeds, indexed by 0, 1, and 2, are used for the 128-unit GNN, a same-width 128-unit pairwise MLP, and a capacity-matched 307-unit pairwise MLP, and every selected allocator is evaluated once on two disjoint 128-mission test batches generated with seeds 314159 and 271828. The corresponding trainable actor parameter counts are 101,762, 19,457, and 101,618, respectively, so the capacity-matched comparison separates graph context from raw parameter count. All three allocators use the same 128-unit centralized value network, preventing critic capacity from changing across the actor-representation ablation. The random and nearest assignment rules are evaluated on the same two 128-episode batches used by the learned allocators. Learned high-level assignments are sampled without replacement from the masked categorical policy using fixed test random streams, whereas low-level actions use the deterministic actor mean; every rollout terminates immediately upon complete coverage. The low-level controller and safety-filter component studies use the predeclared GNN training replicate indexed by 0 and do not select an allocator from the final test results. The low-level critic ablation copies the actor and reward critic of the frozen full-method controller into both a MAPPO branch and a Safe-MAPPO branch, while the Safe-MAPPO-only cost-value head is initialized independently with the predeclared seed 2026. Both branches are then adapted for 100 updates under three paired rollout seeds with identical environments, PPO hyperparameters, update budgets, and execution filters, and every terminal checkpoint is evaluated without test-based selection. This controlled final-adaptation study measures the marginal effect of the safety-cost critic from a common mature controller rather than claiming a from-scratch sample-efficiency comparison. Mission realizations are paired by episode index across methods, including their active targets, initial states, and high-level sampling streams. For learned allocators, reported means first aggregate each 256-mission test replicate and then aggregate the three independent training replicates; uncertainty intervals use a nested percentile bootstrap that resamples training replicates and paired mission indices. Paired GNN-minus-MLP effects use 8000 nested bootstrap draws over common training-replicate and mission indices, while method-level intervals use 4000 draws; prescribed Random and Nearest allocators have no training-seed level and are resampled only over their common test missions.

Let E denote the number of evaluation episodes, let n_e denote the terminal step of episode e, and let $C_e(n_e)$ denote its final coverage. The success rate (SR) and mean completion steps are defined as

$$\mathrm{SR} = \frac{1}{E}\sum_{e=1}^{E} \mathbb{I}\{C_e(n_e) = 1\}\,, \qquad \bar{n} = \frac{1}{E}\sum_{e=1}^{E} n_e. \tag{41}$$

Safety is summarized by the smallest inspector center distance, the smallest signed station keep-out margin, and the raw signed distance from an inspector center to the uninflated station primitives,

$$\begin{aligned} d^{\mathrm{I}}_{\min} &= \min_{e,n}\min_{i<\ell} \|p_{i,e,n} - p_{\ell,e,n}\|_2\,, \quad h^{\mathrm{S}}_{\min} = \min_{e,n}\min_{i,\ell\in\mathcal{O}_{\mathrm{S}}} h^{\mathrm{S}}_{i\ell,e,n}, \\ d^{\mathrm{S,raw}}_{\min} &= \min_{e,n}\min_{i,\ell\in\mathcal{O}_{\mathrm{S}}} d^{\mathrm{S}}_{\ell}(p_{i,e,n}). \end{aligned} \tag{42}$$

Three pairwise clearances accompany the minimum distance:

$$
\begin{aligned}
m_{\min}^{\mathrm{I,phys}} &= d_{\min}^{\mathrm{I}} - 2r_{\mathrm{B}}, \qquad & m_{\min}^{\mathrm{I,CBF}} &= d_{\min}^{\mathrm{I}} - d_{\mathrm{safe}}, \\
m_{\min}^{\mathrm{I,ctrl}} &= d_{\min}^{\mathrm{I}} - \left(d_{\mathrm{safe}} + m_{\mathrm{s}}^{\mathrm{I}}\right). & &
\end{aligned}
\tag{43}
$$

The three quantities measure physical bounding-sphere clearance, nominal CBF margin, and sampled-data control margin, respectively. The sampled-data station control margin is

$$
\widetilde{h}_{\min}^{\mathrm{S}} = \min_{e,n,i,\ell} d_{\ell}^{\mathrm{S}}\left(p_{i,e,n}; m_{\mathrm{s}}^{\mathrm{S}}\right), \tag{44}
$$

while $h_{\min}^{\mathrm{S}}$ measures margin relative to the nominal inflated CBF envelope. The MuJoCo contact count N_{c} measures intersection of the detailed physical collision geometries. The raw metric $d_{\min}^{\mathrm{S,raw}}$ removes CBF inflation and is used to distinguish gross center penetration from activation of the conservative keep-out envelope rather than as a rigid-body contact certificate. Consequently, a small negative keep-out margin indicates use of an elastic CBF relaxation and does not by itself imply physical contact. Every diagnostic evaluation also records the maximum elastic barrier relaxation, the fraction of active QP solves whose relaxation exceeds $\varepsilon_{\mathrm{QP}}$, the numerical convergence rate, and the mean solver iteration count for the translational and attitude filters separately. Compile-excluded filter latency is measured after a warm-up solve and is summarized by its mean and 95th percentile, while MuJoCo contact validation is performed independently at every recorded 0.05 s safety-integration state, including the initial state of each episode. The translational and rotational intervention magnitudes are reported separately as

$$
\begin{aligned}
\bar{\Delta}_a &= \frac{1}{\sum_e n_e N} \sum_{e=1}^{E} \sum_{n=0}^{n_e-1} \sum_{i=1}^{N} \left\| a_{i,e,n,\mathrm{safe}}^{\mathrm{S}} - \bar{a}_{i,e,n}^{\mathrm{S}} \right\|_2, \\
\bar{\Delta}_\tau &= \frac{1}{\sum_e n_e N} \sum_{e=1}^{E} \sum_{n=0}^{n_e-1} \sum_{i=1}^{N} \left\| \tau_{i,e,n,\mathrm{safe}}^{b} - \bar{\tau}_{i,e,n}^{\mathrm{P}} \right\|_2.
\end{aligned}
\tag{45}
$$

The instantaneous pointing error is

$$
\theta_{ij,e,n} = \cos^{-1}\left(\mathrm{clip}_{[-1,1]}\left(b_{i,e,n}^{\top} d_{ij,e,n}\right)\right), \tag{46}
$$

and the inspection condition $a_{\mathrm{vis}} = 0.85$ corresponds to a maximum admissible pointing error of approximately 31.8° at a visit event. Coverage, success rate, completion steps, return, filter intervention, minimum safety margins, pointing error, and MuJoCo contacts form the common evaluation criteria in the following experiments.

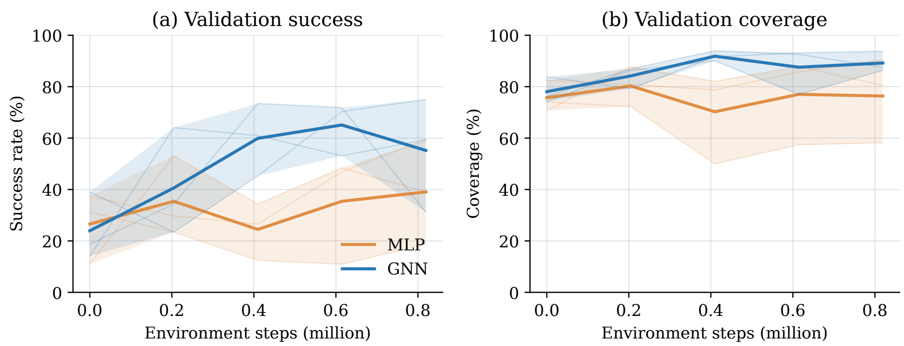

Figure 5: Current-policy learning curves of the GNN-128, MLP-128, and capacity-matched MLP-307 allocators on the fixed 128-mission validation set, where faint lines show independent training seeds, solid lines show their mean, and shaded regions denote pointwise 95% bootstrap confidence intervals; the 1280-step budget is the predeclared checkpoint-selection budget.

4.2. Overall Inspection Performance

Figure 5 evaluates each current stochastic assignment policy before training and after every five closed-loop PPO updates, rather than displaying a cumulative best-so-far statistic as the current policy. Success within the fixed 1280-step selection budget, mission duration, team path length, and the predeclared validation objective expose learning progress that saturated terminal coverage alone would hide. The validation-based checkpoint selection described in Section 4.1 remains independent of both untouched test batches used below.

Figure 6 shows that inspection progress is accumulated through distinct coverage events while both safety filters intervene only when required by the nominal commands. Every visit-event pointing error remains below the 31.8° threshold required to register an inspection target. The aggregate mission, barrier, and detailed-contact statistics over the two untouched test batches are reported together with the component-wise comparisons in Section 4.3.

4.3. Comparisons and Ablation Studies

The comparison protocol is divided into a high-level allocation study and a low-level control-and-safety study so that only one architectural layer is varied at a time. The high-level study fixes the trained Safe-MAPPO controller, the translational CBF, the pointing HOCBF, the masked assignment

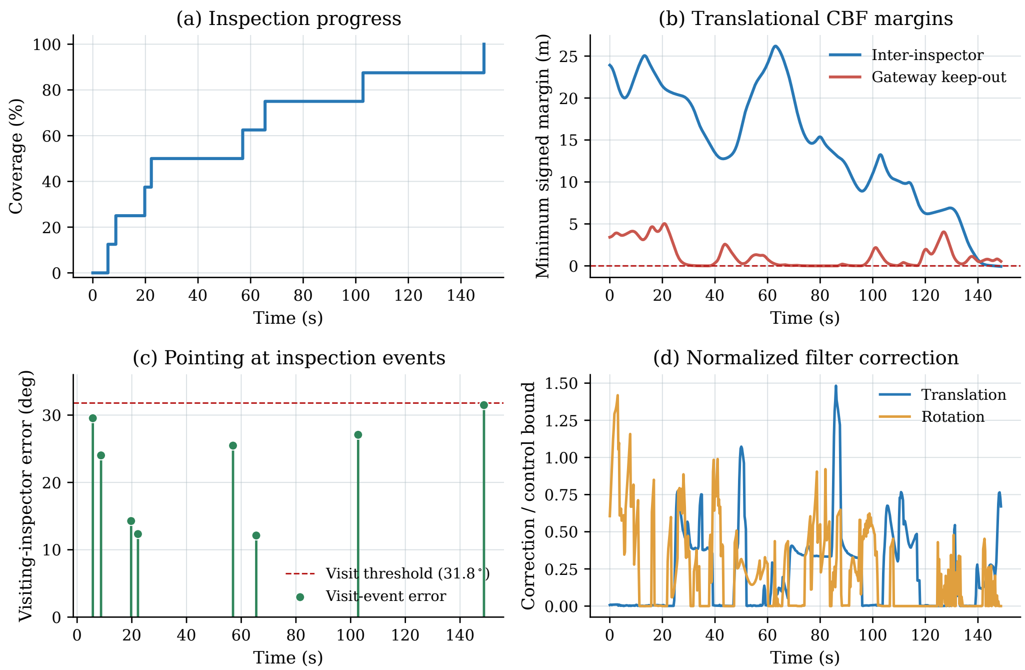

Figure 6: Closed-loop response of the complete hierarchical method for the successful episode whose completion time is closest to the median of a designated 128-mission independent test batch. Translational margins are measured relative to the pairwise and inflated Gateway CBF boundaries, and green markers identify inspectors that trigger coverage while satisfying the pointing threshold. The control-correction panel normalizes translational and rotational interventions by $a_{\max}$ and $\tau_{\max}$, respectively, and treats corrections at or below the QP tolerance as zero.

decoder, and all environment settings. Only the assignment-utility source changes across the compared methods.

The random-allocation baseline samples a feasible assignment uniformly without replacement and therefore retains the same goal-conditioned low-level interface without optimizing the high-level decisions. The nearest-allocation baseline sets the edge score to $-\|p_i(t_k) - p_j^{\mathrm{v}}\|_2$ and applies the same masked greedy decoder as the learned allocators. The pairwise MLP independently maps each edge feature ξ_{ij}^k to a utility, whereas the GNN uses the inspector and target message-passing summaries in Eq. (18) before decoding the utility. The same-width comparison uses 128 hidden units in both networks, and the capacity control increases the pairwise MLP width to 307 so that its parameter count matches the GNN within 0.15%. All

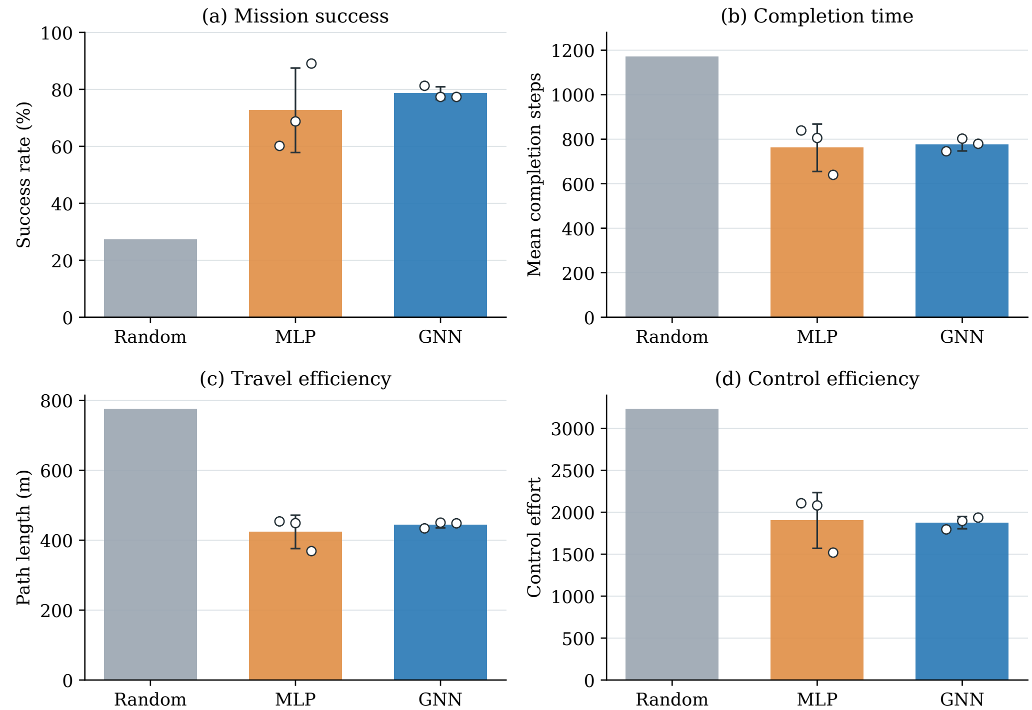

Figure 7: Matched closed-loop comparison of Random, Nearest, MLP-128, capacity-matched MLP-307, and GNN-128 task allocation. Curves and shaded bands report method means and 95% bootstrap intervals, while lower-row markers and whiskers report means and 95% bootstrap intervals and the unfilled points identify individual learned-policy training replicates. Every method is evaluated on the same two independent 128-mission test batches, and lower completion steps, path length, and control effort are preferable.

learned allocators use identical pairwise inputs, training reward, option horizon, optimizer, and update budget. Neither learned architecture receives a nearest-distance prior, nearest-target label, or behavior-cloning target, so the MLP–GNN comparison isolates graph message passing under a common pure reinforcement-learning protocol.

Figure 7 shows that random assignment attains only 22.66% success, confirming that feasible but uncoordinated target selection is insufficient for the randomized eight-target missions. The GNN attains 79.69% $\pm$ 3.58% success across three closed-loop training seeds, compared with 70.05% $\pm$ 14.07% for the pairwise MLP, thereby improving mean success by 9.64 percentage points and reducing the success-rate standard deviation by 74.6%. The cor-

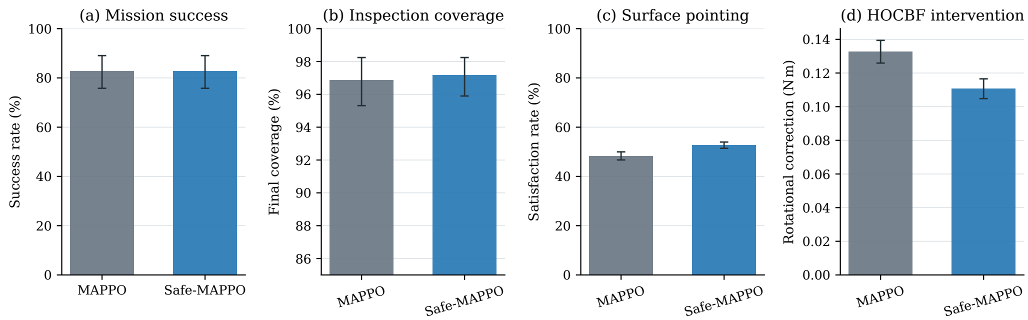

Figure 8: Matched MAPPO and Safe-MAPPO evaluation with the same selected GNN allocator and CBF/HOCBF execution filters, where lower-row error bars denote nested-bootstrap 95% confidence intervals over three adaptation replicates and paired missions from two common 128-mission test batches, and unfilled points denote replicate means.

responding mean coverage values are 95.90% ± 1.13% and 94.08% ± 1.81%, while the GNN reduces mean completion time from 791.2 to 762.4 steps and mean control effort from 1968.4 to 1844.0. The GNN mean path is 439.3 m compared with 432.8 m for the MLP, but its path-length standard deviation decreases from 44.6 to 6.8 m. The nearest heuristic is retained as a classical reference and reaches 81.25% success with 604.6 mean steps and a 319.5 m mean path, but it is not used as the learned-architecture ablation target. The learned comparison therefore establishes that relational message passing improves PPO-based assignment success and seed robustness without receiving the heuristic as a prior.

Figure 8 isolates the low-level safety-aware critic by keeping the high-level assignments and execution filters fixed. MAPPO and Safe-MAPPO both attain 82.81% success, while Safe-MAPPO increases mean coverage from 96.88% to 97.17% and surface-pointing satisfaction from 48.32% to 52.68%. Safe-MAPPO also reduces the mean rotational HOCBF correction from 0.1327 to 0.1107 N m and the mean control effort per executed step from 2.826 to 2.385, although its mean completion time increases from 564.8 to 749.9 steps. Both filtered controllers produce zero MuJoCo contacts over all 128 trajectories, attributing this behavior to the common execution filters. The safety-aware controller principally improves pointing quality while lowering per-step control demand.

The safety-filter ablation then fixes the Safe-MAPPO controller, refined GNN allocator, target sets, initial states, and nominal pointing guidance.

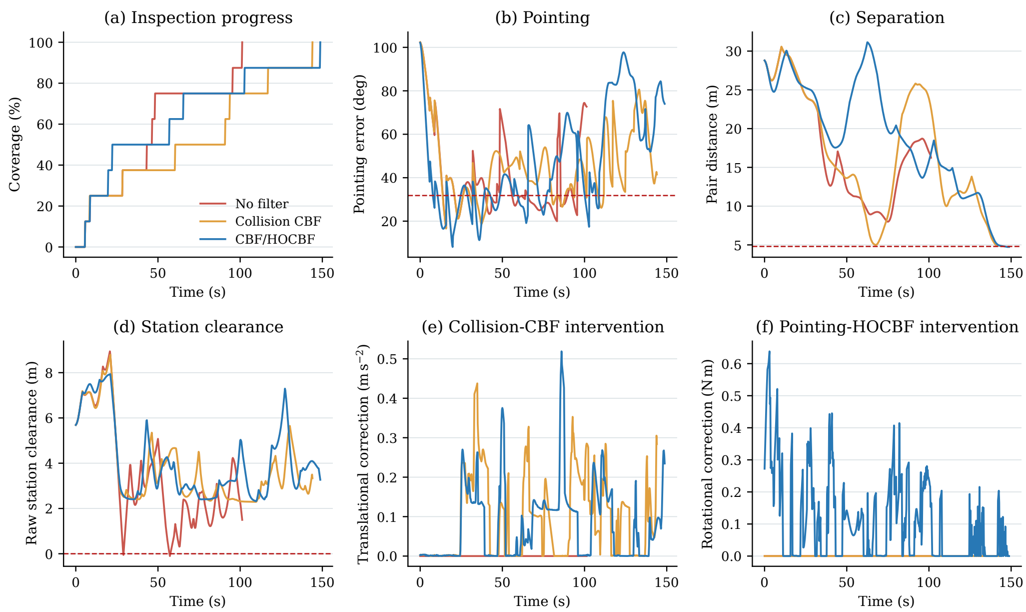

Figure 9: Matched closed-loop histories for the no-filter, collision-CBF, and complete CBF/HOCBF variants under the same target set and initial state. The displayed mission is the successful complete-method episode whose completion time is closest to the median of the 128-episode batch, and red dashed lines denote the pointing, pairwise-separation, and raw station-clearance boundaries.

The no-filter variant applies the nominal control input directly, the collision-CBF variant applies only the translational CBF-QP, and the complete variant additionally applies the pointing HOCBF-QP.

Figure 9 resolves the temporal behavior hidden by terminal aggregate metrics. The no-filter trajectory completes rapidly but penetrates the raw Gateway collision geometry, whereas both collision-filtered trajectories remain on the positive side of the raw station boundary. The attitude HOCBF produces its largest corrections during initial acquisition and target transitions, and its pointing-error trace reflects repeated reorientation toward the surface point associated with each new assignment. The three trajectories have different termination times because the filters alter the closed-loop motion rather than merely supplying diagnostic constraints.

Figure 10 shows why the unfiltered 100% nominal success rate is not a valid safe solution: its minimum pairwise distance is 0.164 m, its raw station clearance reaches −3.164 m, and every episode contains a MuJoCo contact,

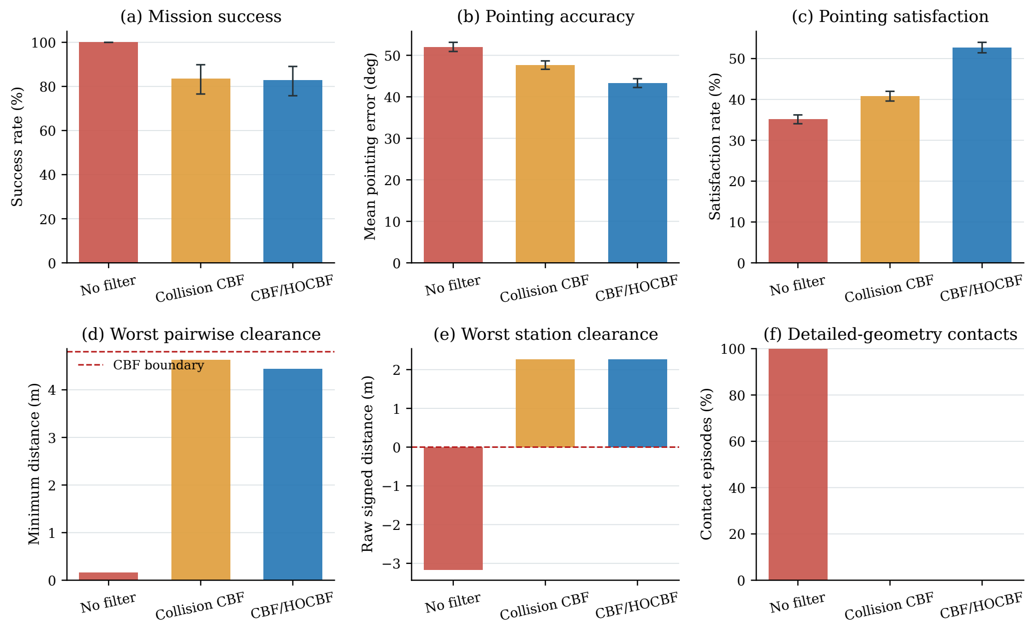

Figure 10: Matched execution-filter ablation over two common 128-mission test batches. Error bars in panels (b) and (c) denote bootstrap 95% confidence intervals, panels (d) and (e) report the worst clearance over all trajectories, and panel (f) gives the episode fraction with a detailed-geometry MuJoCo contact at any safety integration state.

totaling 44,897 contacts over 18,391 contact steps. The collision CBF eliminates all detailed-geometry contacts, raises the worst raw station clearance to 2.269 m, and maintains a worst pair distance of 4.632 m. The residual difference from the nominal 4.8 m pairwise boundary is admitted by the elastic QP relaxation, while the positive raw clearance and zero MuJoCo contacts confirm physical separation in the detailed model. Adding the pointing HOCBF changes success from 83.59% to 82.81%, reduces mean pointing error from 47.64° to 43.30°, and raises pointing satisfaction from 40.78% to 52.68% without introducing any contact.

Taken together, the allocation and control results support an explicit hierarchy because the GNN handles slow combinatorial service decisions while the shared controller learns reusable fast rigid-body motion. The masked decoder removes duplicate open-target assignments without providing heuristic labels, and freezing the converged controller prevents a changing low-level transition model during allocator training. The GNN result should be interpreted as a relational-representation advantage over matched learned MLPs rather than universal dominance over nearest assignment, whose geometric bias is strong when targets are homogeneous and travel distance dominates.

Safe-MAPPO and the execution filters have complementary roles: the safety-cost critic shapes nominal policy updates, whereas the CBF/HOCBF QPs project every executed command and the MuJoCo audit checks the detailed geometry independently. Elastic variables preserve QP feasibility under conflicting barriers and actuator bounds, but nonzero relaxation precludes an unconditional forward-invariance claim. Positive raw clearance, zero contacts, barrier margins, and relaxation statistics therefore provide empirical safety evidence for the tested mission distribution without overstating formal certification.

The present evidence is limited to three inspectors, a finite viewpoint catalog, known relative states, deterministic station geometry, and simulation-based contact validation. The sequential decoder guarantees nonduplicate assignments but not globally optimal matching, and the attitude-independent inspector sphere with capsule/OBB station geometry remains conservative. Larger teams, continuous image-utility coverage [6], uncertain dynamics and perception, adaptive barrier margins [25], actuator allocation, and hardware-in-the-loop validation remain future extensions.

5. Conclusion

This paper presented a safe hierarchical MARL method for close-proximity inspection by multiple spacecraft. The method combines a GNN high-level allocator with a goal-conditioned Safe-MAPPO controller on SE(3). CBF and HOCBF action filters enforce station clearance, inter-inspector separation, and surface pointing during execution. The hierarchy separates discrete relational coordination from continuous rigid-body execution, while the safety layer exposes constraint behavior independently of the learned mission reward. Matched closed-loop comparisons and ablations demonstrate the roles of graph message passing, centralized safety-aware training, and explicit action filtering, and detailed MuJoCo contact checks complement the analytic barrier metrics. The resulting formulation provides a reproducible basis for studying learned task allocation and safety-critical execution within the same spacecraft inspection mission.

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